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ILSA PROJECT

Train Protection- VF.CONCEPT SFRS

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00	25/02/2021		First emission
01	30/06/2021	Modified: Vigilance flow chart (Figure 2) 2F_09.02.Zefiro-ES.CONCEPT.140 2F_09.02.Zefiro-ES.CONCEPT.141 2F_09.02.Zefiro-ES.CONCEPT.142 2F_09.02.Zefiro-ES.CONCEPT.143 2F_09.02.Zefiro-ES.CONCEPT.144 2F_09.02.Zefiro-ES.CONCEPT.145 2F_09.02.Zefiro-ES.CONCEPT.146 2F_09.02.Zefiro-ES.CONCEPT.150 2F_09.02.Zefiro-ES.CONCEPT.152 2F_09.02-Zefiro-ES.CONCEPT.155 2F_09.02.Zefiro-ES.CONCEPT.185 (only SIL) 2F_09.02.Zefiro-ES.CONCEPT.186 2F_09.02.Zefiro-ES.CONCEPT.187 2F_09.02.Zefiro-ES.CONCEPT.197 2F_09.02.Zefiro-ES.CONCEPT.200 (allocation) 2F_09.02.Zefiro-ES.CONCEPT.199 (only SIL) 2F_09.02.Zefiro-ES.CONCEPT.200 (only SIL) 2F_09.02.Zefiro-ES.CONCEPT.228 (only SIL) 2F_09.02.Zefiro-ES.CONCEPT.242 (only SIL) 2F_09.02.Zefiro-ES.CONCEPT.249 (only SIL) 2F_09.02.Zefiro-ES.CONCEPT.282 (allocation) 2F_09.02-Zefiro-ES.CONCEPT.283 2F_09.02-Zefiro-ES.CONCEPT.316 2F_09.02-Zefiro-ES.CONCEPT.317 2F_09.02-Zefiro-ES.CONCEPT.318 2F_09.02-Zefiro-ES.CONCEPT.319 (also moved to par 0) 2F_09.02-Zefiro-ES.CONCEPT.331 2F_09.02-Zefiro-ES.CONCEPT.354 2F_09.02-Zefiro-ES.CONCEPT.355 2F_09.02-Zefiro-ES.CONCEPT.356 2F_09.02-Zefiro-ES.CONCEPT.357 2F_09.02-Zefiro-ES.CONCEPT.358 2F_09.02.Zefiro-ES.CONCEPT.373 (allocation) 2F_09.02-Zefiro-ES.CONCEPT.435 (allocation) 2F_09.02-Zefiro-ES.CONCEPT.553 (allocation) 2F_09.02-Zefiro-ES.CONCEPT.439 2F_09.02-Zefiro-ES.CONCEPT.440	Second emission

Revision	Date	Changes	Reason
		2F_09.02-Zefiro-ES.CONCEPT.441 updated references: RD 1, RD 5, RD 8 Added: Clarification in 3.1.2.20.1 2F_09.02-Zefiro-ES.CONCEPT.567 2F_09.02-Zefiro-ES.CONCEPT.568 2F_09.02-Zefiro-ES.CONCEPT.569 2F_09.02-Zefiro-ES.CONCEPT.570 2F_09.02-Zefiro-ES.CONCEPT.571 2F_09.02-Zefiro-ES.CONCEPT.572 2F_09.02-Zefiro-ES.CONCEPT.573 2F_09.02-Zefiro-ES.CONCEPT.574 2F_09.02-Zefiro-ES.CONCEPT.575 2F_09.02-Zefiro-ES.CONCEPT.576 2F_09.02-Zefiro-ES.CONCEPT.577 2F_09.02-Zefiro-ES.CONCEPT.578 2F_09.02-Zefiro-ES.CONCEPT.579 2F_09.02-Zefiro-ES.CONCEPT.580 2F_09.02-Zefiro-ES.CONCEPT.581 2F_09.02-Zefiro-ES.CONCEPT.582 2F_09.02-Zefiro-ES.CONCEPT.583 2F_09.02-Zefiro-ES.CONCEPT.584 2F_09.02-Zefiro-ES.CONCEPT.585 2F_09.02-Zefiro-ES.CONCEPT.586 2F_09.02-Zefiro-ES.CONCEPT.587 2F_09.02-Zefiro-ES.CONCEPT.588 2F_09.02-Zefiro-ES.CONCEPT.589 2F_09.02-Zefiro-ES.CONCEPT.590 2F_09.02-Zefiro-ES.CONCEPT.591 2F_09.02-Zefiro-ES.CONCEPT.592 2F_09.02-Zefiro-ES.CONCEPT.593 2F_09.02-Zefiro-ES.CONCEPT.594 2F_09.02-Zefiro-ES.CONCEPT.595 2F_09.02-Zefiro-ES.CONCEPT.596 2F_09.02-Zefiro-ES.CONCEPT.597 2F_09.02-Zefiro-ES.CONCEPT.598 2F_09.02-Zefiro-ES.CONCEPT.599 2F_09.02-Zefiro-ES.CONCEPT.600 2F_09.02-Zefiro-ES.CONCEPT.601 2F_09.02-Zefiro-ES.CONCEPT.602 2F_09.02-Zefiro-ES.CONCEPT.603 2F_09.02-Zefiro-ES.CONCEPT.604 2F_09.02-Zefiro-ES.CONCEPT.605	

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		2F_09.02-Zefiro-ES.CONCEPT.606 2F_09.02-Zefiro-ES.CONCEPT.607 2F_09.02-Zefiro-ES.CONCEPT.621 2F_09.02-Zefiro-ES.CONCEPT.622 2F_09.02-Zefiro-ES.CONCEPT.623 2F_09.02-Zefiro-ES.CONCEPT.624 2F_09.02-Zefiro-ES.CONCEPT.625 2F_09.02-Zefiro-ES.CONCEPT.627 2F_09.02-Zefiro-ES.CONCEPT.644 Deleted: 2F_09.02-Zefiro-ES.CONCEPT.457 2F_09.02-Zefiro-ES.CONCEPT.458	
02	2409/2021	Modified requirements: 2F_09.02-Zefiro-ES.CONCEPT.241 2F_09.02-Zefiro-ES.CONCEPT.242 2F_09.02-Zefiro-ES.CONCEPT.582 2F_09.02-Zefiro-ES.CONCEPT.583 2F_09.02-Zefiro-ES.CONCEPT.439 2F_09.02-Zefiro-ES.CONCEPT.440 2F_09.02-Zefiro-ES.CONCEPT.583* 2F_09.02-Zefiro-ES.CONCEPT.593* 2F_09.02-Zefiro-ES.CONCEPT.601* Added requirements: 2F_09.02-Zefiro-ES.CONCEPT.645 2F_09.02-Zefiro-ES.CONCEPT.646 2F_09.02-Zefiro-ES.CONCEPT.647 2F_09.02-Zefiro-ES.CONCEPT.648 2F_09.02-Zefiro-ES.CONCEPT.649 2F_09.02-Zefiro-ES.CONCEPT.650 2F_09.02-Zefiro-ES.CONCEPT.653 2F_09.02-Zefiro-ES.CONCEPT.654 2F_09.02-Zefiro-ES.CONCEPT.655 2F_09.02-Zefiro-ES.CONCEPT.656 2F_09.02-Zefiro-ES.CONCEPT.657 2F_09.02-Zefiro-ES.CONCEPT.658 2F_09.02-Zefiro-ES.CONCEPT.659 2F_09.02-Zefiro-ES.CONCEPT.660 2F_09.02-Zefiro-ES.CONCEPT.661 2F_09.02-Zefiro-ES.CONCEPT.662 2F_09.02-Zefiro-ES.CONCEPT.663 2F_09.02-Zefiro-ES.CONCEPT.664 2F_09.02-Zefiro-ES.CONCEPT.665	Third Emission: - Define DVD_TEST_RESULT= ABORTED and added in Table 2: DVD test result - Added requirements for TCO command application during EB and SB. - Added functional requirements for STM panto and MPS management - Added requirements for diagnostic purpose - *: Modified requirements about LZB track conditions management by TCMS in order to manage the missing update of LZB wayside configuration

Revision	Date	Changes	Reason
		2F_09.02-Zefiro-ES.CONCEPT.666 Deleted: 2F_09.02-Zefiro-ES.CONCEPT.576 Reference: Update revision of RD 1 Added RD 9	
03	16/11/2021	Modified: 3.1.2.16.1.1 (clarification) Modified requirements. 2F_09.02-Zefiro-ES.CONCEPT.139 2F_09.02-Zefiro-ES.CONCEPT.646 (clarification) 2F_09.02-Zefiro-ES.CONCEPT.203 (only allocation) 2F_09.02-Zefiro-ES.CONCEPT.228 (clarification) 2F_09.02-Zefiro-ES.CONCEPT.241 (formal) 2F_09.02-Zefiro-ES.CONCEPT.244 2F_09.02-Zefiro-ES.CONCEPT.260 (formal) 2F_09.02-Zefiro-ES.CONCEPT.656 (formal) 2F_09.02-Zefiro-ES.CONCEPT.436 (only allocation) 2F_09.02-Zefiro-ES.CONCEPT.437 (only allocation) 2F_09.02-Zefiro-ES.CONCEPT.440 (clarification) Added requirements: 2F_09.02-Zefiro-ES.CONCEPT.669 2F_09.02-Zefiro-ES.CONCEPT.670 2F_09.02-Zefiro-ES.CONCEPT.671 2F_09.02-Zefiro-ES.CONCEPT.672 2F_09.02-Zefiro-ES.CONCEPT.673 2F_09.02-Zefiro-ES.CONCEPT.674 2F_09.02-Zefiro-ES.CONCEPT.675 2F_09.02-Zefiro-ES.CONCEPT.676 2F_09.02-Zefiro-ES.CONCEPT.677 2F_09.02-Zefiro-ES.CONCEPT.678 2F_09.02-Zefiro-ES.CONCEPT.679 2F_09.02-Zefiro-ES.CONCEPT.695 2F_09.02-Zefiro-ES.CONCEPT.696 2F_09.02-Zefiro-ES.CONCEPT.697 2F_09.02-Zefiro-ES.CONCEPT.698 2F_09.02-Zefiro-ES.CONCEPT.699 2F_09.02-Zefiro-ES.CONCEPT.700	Fourth emission: Added functionality about: - ASFA automatic Power ON/OFF with cab status (CR209748) - Automatic Vigilance activation/inhibition with cab status - Defined the values of the parameters to be used by TCMS and ETCS during track condition management - Added requirements for diagnostic messages releated to bullets above and MCB of ATP systems Corrected minor formal error

Revision	Date	Changes	Reason
		2F_09.02-Zefiro-ES.CONCEPT.701 2F_09.02-Zefiro-ES.CONCEPT.702 Reference: Updated version of RD 1 and RD 8 Removed brake SFRS reference	
04	18/02/2022	Modified: 2F_09.02-Zefiro-ES.CONCEPT.154 (SIL) 2F_09.02-Zefiro-ES.CONCEPT.155 (clarification) 2F_09.02-Zefiro-ES.CONCEPT.156 (SIL) 2F_09.02-Zefiro-ES.CONCEPT.260 (clarification) 2F_09.02-Zefiro-ES.CONCEPT.281 (clarification) 2F_09.02-Zefiro-ES.CONCEPT.282 (clarification) 2F_09.02-Zefiro-ES.CONCEPT.283 (note) 2F_09.02-Zefiro-ES.CONCEPT.284 (clarification) 2F_09.02-Zefiro-ES.CONCEPT.285 (clarification) 2F_09.02-Zefiro-ES.CONCEPT.621 2F_09.02-Zefiro-ES.CONCEPT.649 (SIL) Added Requirements: 2F_09.02-Zefiro-ES.CONCEPT.704 2F_09.02-Zefiro-ES.CONCEPT.705 2F_09.02-Zefiro-ES.CONCEPT.706 2F_09.02-Zefiro-ES.CONCEPT.707 2F_09.02-Zefiro-ES.CONCEPT.708 2F_09.02-Zefiro-ES.CONCEPT.709 2F_09.02-Zefiro-ES.CONCEPT.710	Fifth emission: - Added requirement about TCO by STM LZB. - Introduction of switchable data entry. - Minor formal correction

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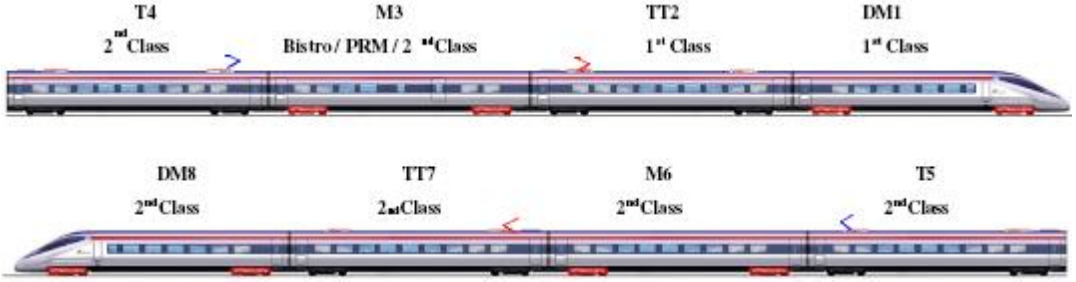
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1 Introduction

This document is a function specification of ATP system for V300ZEFIRO ILSA project.

1.1 Train Configuration

Each V300ZEFIRO ILSA train consists of eight cars in a fixed formation including a driver's cab at each end:



- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

1.2 Definitions and abbreviations

Acronym	Meaning
ASFA	Anuncio de Señales y Frenado Automático
DVD	Driver's Vigilance Device
DVD	Driver's Safety Device
ETCS	European Rail Traffic Management System/European Train Control System
LZB	Linienzugbeeinflussung
SABOB	SAfety BOard B (included in teloc 3000 system)

2 System Overview

2.1 System Configuration

The following figure shows the architecture of the ATP system:

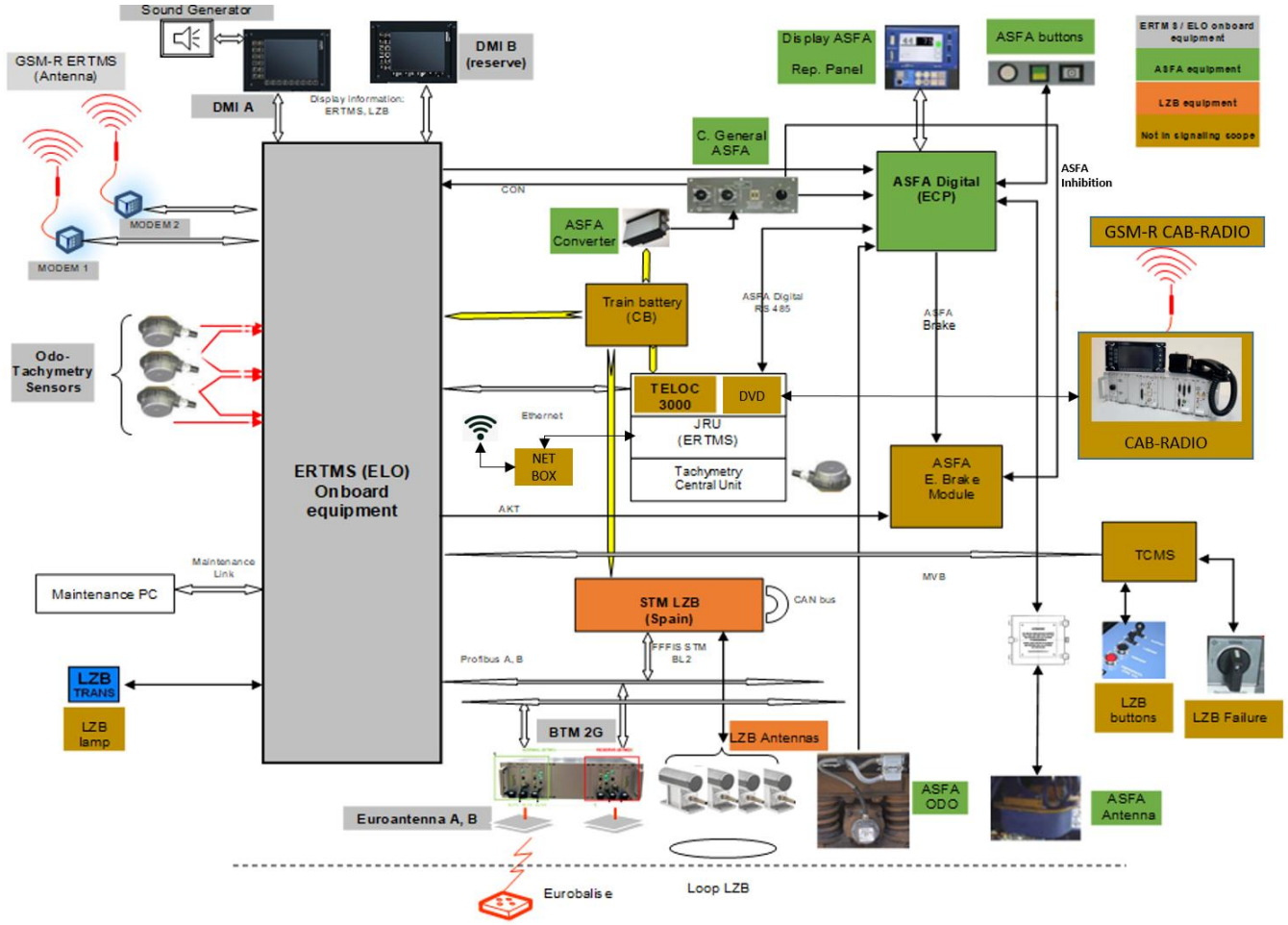


Figure 1: ATP configuration

ATP equipment:

- ETCS BL3 (with LZB integrated)
- ASFA system (independent form ETCS)
- Cab Radio eRace2
- TELOC3000/JRU

3 Functional requirements

3.1 Train Protection

3.1.1 Driver´s Safety Device (DVD)

The vigilance device is integrated into the JRU/TELOC and therefore the DVD controls (pushbuttons on the braking/traction lever and pedal) are managed by the JRU/TELOC system control unit through SABOB board.

3.1.1.1 Driver´s Vigilance Device (DVD)

3.1.1.1.1 Function Design

The vigilance device monitors the presence and vigilance of the driver through the activation of the controls in the cabin.
In case the driver vigilance or presence is not confirmed the DVD switch on an optical warning, then acoustical warning and in the end it applies the emergency brake. The event is then recorded by the train recorder and a message is sent to the ground by the cab radio if the EB is not released.
The JRU/TELOC system will use one digital input (voltage driven) to monitor the driver vigilance, called “primary chain” in which are connected braking/traction lever’s pushbutton and the pedal.
Refer to the following picture.

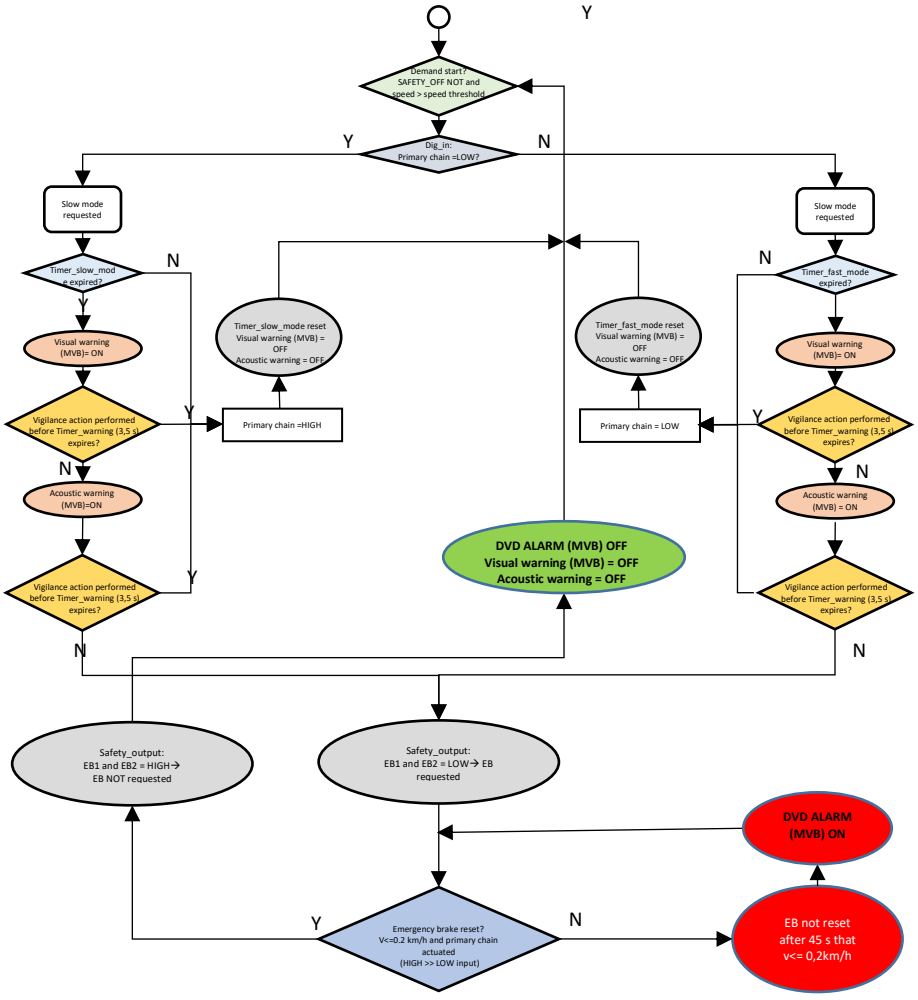


Figure 2: Vigilance flow chart

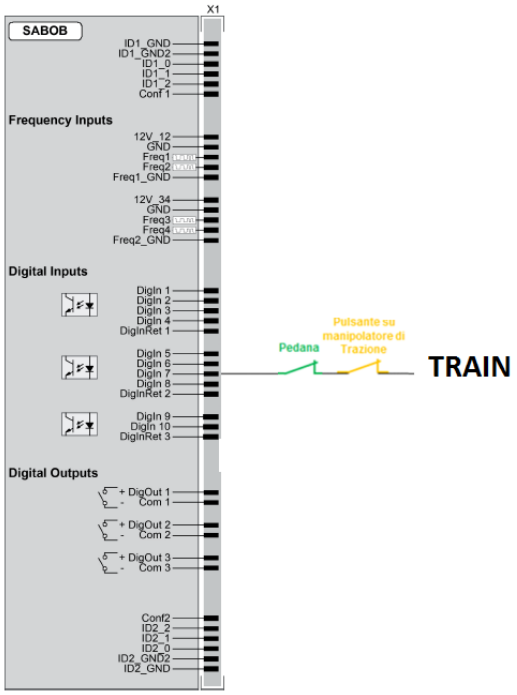


Figure 3: Vigilance device scheme

3.1.1.1.1.1 Normal condition

The driver confirms its presence and vigilance through the pedal or the pushbutton on the traction lever: there is no buzzer and no brake application.
The driver does not confirm its presence or vigilance: the DVD activates the visual and then acoustic warning and if there is still no action from the driver it applies the emergency brake.

When the primary channel modifies its status, the timer shall be reset.
When the primary channel is not actuated the fast mode (vigilance action is requested every 2,5 s) of DVD function shall be requested.
When the primary channel is actuated the slow mode (vigilance action is requested every 30 s) of DVD function shall be requested

3.1.1.1.1.2 Failure cases

A. DVD applies the brake when not needed => The train is stopped
B. DVD fails to apply the brake when commanded => The train is not stopped when needed
C. Primary channel fails during trip

3.1.1.1.1.3 Failure detection

A. Failure self-evident
B. Failure self-evident
C. DVD EB is activated

3.1.1.1.1.4 Failure handling

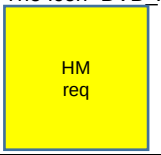
A. The DVD shall be isolated and the system shall be repaired during maintenance
B. The DVD shall be isolated and the system shall be repaired during maintenance
C. The DVD shall be isolated and the system shall be repaired during maintenance

3.1.1.1.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.105
Description	The driving personnel shall confirm presence and vigilance by activating the safety pedal or pushbuttons.
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.137	The status of the safety pedal AND B/T lever's pushbutton (operated/not operated) shall be provided to the DVD (JRU/TELOC) on the primary chain.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2

ID	2F_09.02.Zefiro-Europe.TRS.145
Description	Emergency brake shall be applied if the driver does not confirm the presence after first an optical warning and after that an acoustic warning.
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.139	The vigilance functionality shall be de-activated when the desk is disabled acting on DVD_INHIBITION input of the TELOC	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.140	If the primary chain is not actuated, the fast mode of DVD shall be required: performing an action on primary chain within 2,5 s (Fast mode Timer is configurable)	Derived Requirement	4S_09.06.06.-.Data Recorder - Juridical Recorder		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.141	If the primary chain is actuated, the slow mode of DVD shall be required: performing an action on primary chain within 30 s (Slow mode Timer is configurable)	Derived Requirement	4S_09.06.06.-.Data Recorder - Juridical Recorder		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.142	If no action is performed before the timer (slow or fast mode) is expired, a visual warning shall be set on MVB bus DVD_Visual_Warning=1	Derived Requirement	4S_09.06.06.-.Data Recorder - Juridical Recorder		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.143	When TCMS reads DVD_Visual_Warning=1 it shall show on IDU the icon for DVD action requested.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.567	The Icon "DVD_Action Requested" shall be like: 	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_09.02-Zefiro-ES.CONCEPT.144	If no action is performed after 3,5 s (configurable timer) after visual warning, teloc system shall set the acoustic warning on MVB bus: DVD_acoustic_warning=1	Derived Requirement	4S_09.06.06.-.Data Recorder - Juridical Recorder		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.145	When TCMS reads DVD_acoustic_warning=1 it shall command a dedicated buzzer in the cabin.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.568	If no action is performed after 3,5 s (configurable timer) after buzzer activation request, teloc system shall command the Emergency brake.	Derived Requirement	4S_09.06.06.-.Data Recorder - Juridical Recorder		SIL 2
2F_09.02-Zefiro-ES.CONCEPT.146	Emergency Brake requested by DVD shall de-energize Emergency Brake Loop"	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_09.02-Zefiro-ES.CONCEPT.147	When the Emergency brake is triggered, for safety reason, the train shall reach standstill condition, before the EB can be released	Derived Requirement	4S_09.06.06.-.Data Recorder - Juridical Recorder		SIL 2
2F_09.02-Zefiro-ES.CONCEPT.148	In order to release the DVD_EB_INTERVENTION, the driver shall perform the following actions at standstill. The actions are reported below: - If the primary chain is actuated during when standstill condition is reached, the actuated organ shall be released and then pedal or pushbutton shall be pressed again - If the primary chain is not actuated when standstill is reached, one organ shall be pressed	Derived Requirement	4S_09.06.06.-.Data Recorder - Juridical Recorder		SIL 2
2F_09.02-Zefiro-ES.CONCEPT.149	The JRU shall close the EB_command outputs at start up.	Derived Requirement	4S_09.06.06.-.Data Recorder - Juridical Recorder		SIL0
2F_09.02-Zefiro-ES.CONCEPT.150	TCMS shall acquire the status of DVD_EB commands and set an event when an incoherence is detected	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.02-Zefiro-ES.CONCEPT.151	TCMS shall show on IDU a message, when the DVD_EB_intervention is requested	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.02-Zefiro-ES.CONCEPT.152	The JRU/TELOC shall record the DVD EB intervention.	Derived Requirement	4S_09.06.06.-.Data Recorder - Juridical Recorder		SIL 0

ID	2F_09.02.Zefiro-Europe.TRS.147
Description	The DVD shall be interfaced with the Cab radio through MVB for the generation of the GSM-R alarm in case of DVD intervention
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.154	The DVD shall provide through MVB to the cab radio the DVD_ALARM signal.	Derived Requirement	4S_09.06.06.-.Data Recorder - Juridical Recorder		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.155	Teloc shall send DVD_ALARM=1, 45 s (configurable timer) after the standstill condition is reached and the EB is not released by the driver yet.	Derived Requirement	4S_09.06.06.-.Data Recorder - Juridical Recorder		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.704	When the driver release the EB acting on vigilance device, all alarm and warning signal (MVB) shall be reset	Derived Requirement	4S_09.06.06.-.Data Recorder - Juridical Recorder		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.569	When TCMS reads DVD_ALARM = 1 (telegram 0x04FF), it shall sow on IDU the DVD_ALARM icon	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.570	The Icon “DVD_ALARM” shall be like: 	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.571	The Icon shall be overlapped on the other Icon already present in the foreseen icon area (first hypothesis: overlap on automatic “PCF/POC unavailable” icon) with the following priority: 1: HM_ALARM 2: HM_action_requested 3: PCF/POC	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.156	The Cab radio shall send the alarm message when receive the DVD_ALARM signal	Derived Requirement	4S_10.01.-.Train Radio System		SIL 0

ID	2F_09.02.Zefiro-Europe.TRS.148
Description	The DVD shall get digital input to detect the driver desk status
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.158	The status of the cab (activated/not activated) shall be provided to the DVD.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2

3.1.1.2 Driver´s Vigilance Device test

3.1.1.2.1 Function design

3.1.1.2.1.1 Normal condition

The test of DVD shall be performed before data entry in order to verify the availability of DVD functionality.
At the end of the test, the operator/driver is informed of the result. In case of “Partially OK” result, the operator/driver is informed about which device is not working properly in order to not use it during the trip.

3.1.1.2.1.2 Failure cases

A. Primary channel devices both fault in NC position
B. One of primary channel device is fault in open condition
C. Fault of one of primary channel device during mission

3.1.1.2.1.3 Failure detection

A. Failure detected during DVD test
B. Failure detected during DVD test
C. DVD emergency brake

3.1.1.2.1.4 Failure handling

A. DVD shall be isolated, and the fault shall be repaired asap
B. DVD shall be isolated, and the fault shall be repaired asap
C. DVD shall be isolated, and the fault shall be repaired asap

3.1.1.2.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.254
Description	The DVD test shall be started manually by the driver in the active cab.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.171	The diagnostic display shall foresee a touch key to command the DVD test.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_09.02-Zefiro-ES.CONCEPT.172	When the driver pushes the relevant soft key on the IDU, the TCMS shall send DVD test message to the JRU though MVB.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.173	The vigilance test shall be performed accordingly to Table 1 below	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02-Zefiro-ES.CONCEPT.174	The vigilance test shall be performed accordingly to Table 1 below	Derived Requirement	4S_09.06.06.-.Data Recorder - Juridical Recorder		SIL0
2F_09.02-Zefiro-ES.CONCEPT.572	The partial test about pedal (step 3.d) and braking/traction lever (step 3.f) pushbuttons shall be considered valid only if the related device is pressed (stable state) for at least 2,5 s	Derived Requirement	4S_09.06.06.-.Data Recorder - Juridical Recorder		SIL0

DVD test procedure	
1.	Driver select DVD test on IDU. TCMS shall read the status of DVD inhibition selector and DESK_status.
2.	If function is inhibited or DESK_STATUS=CLOSED, the DVD_TEST SW button shall be not pressable.
3.	If the vigilance functionality is active (NOT ISOLATED):
3.a)	TCMS shows on IDU the message:" Release all vigilance devices" for 5 seconds
3.b)	TCMS send through MVB the variable "DVD_TEST_ACTIVATION" = activated when the 5 seconds in the previous step are elapsed
3.c)	TCMS informs the driver through IDU:" Press pedal and release it after about 5 s"
3.d)	JRU should read released→ actuated transition of primary channel within 20 s and its successive releasing (within 10 s) (DVD_Pedal_test =OK). if no, JRU send to TCMS "DVD_Pedal_test =NOT_OK". TCMS shall use this information to inform the driver at the end of the test.
3.e)	TCMS record the result and then inform the driver through IDU:" Press traction lever's pushbutton and release it after about 5 s".
3.f)	JRU should read released→ actuated of primary channel within 20 s and its successive releasing (within 10 s) (DVD_TL_button_test =OK). if no, JRU send to TCMS "DVD_TL_button_test =NOT_OK".
4)	TCMS shall send the Variable: DVD_EB_TEST_ACTIVATION = activated to the JRU
4.a)	When the JRU receive the signal above, it shall command DVD_EB_COMMAND_OUTPUT: REQUESTED within 1s
4.b)	TCMS shall detect the status of the DVD_EB_COMMAND_OUTPUT= requested within 10 s and shall send the Variable: DVD_EB_TEST_ACTIVATION = NOT_activated to the JRU If no TCMS shall consider DVD_EB_TEST_RESULT= NOT OK.

DVD test procedure	
4.c)	When JRU receive the signal DVD_EB_TEST_ACTIVATION = NOT_activated, it shall command DVD_EB_COMMAND_OUTPUT =NOT requested JRU send to TCMS: DVD_EB_TEST_RESULT =OK
4.d)	TCMS shall detect the status of the DVD_EB_COMMAND_OUTPUT = not requested within 10 s If no TCMS shall consider DVD_EB_TEST_RESULT= NOT OK
5.	TCMS send through MVB the variable “DVD_TEST_ACTIVATION” =NOT activated → END OF TEST PROCEDURE

Table 1: DVD test procedure

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.176	Basis on the test result of DVD function, defined in Table 2 below, TCMS shall show on IDU as follows: - NOT_OK → TCMS shall sow on IDU: “Vigilance unavailable. Isolation is requested” - PARTIALLY OK → TCMS shall show on IDU: The %(name of faulty device) is not available for vigilance function” - OK →TCMS shall sow on IDU: “Vigilance test are OK”	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_09.02-Zefiro-ES.CONCEPT.645	If the DVD test cannot be performed (i.e. TELOC/SABOB is faulty, DVD INHIBITION is actuated or incoherent), TCMS shall set the DVD test result as “ABORTED”	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_09.02-Zefiro-ES.CONCEPT.573	For each final result, TCMS shall set an event.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

DVD_Pedal_test	OK	OK	NOT_OK	NOT_OK	x	N\A
DVD_TL_button_test	OK	NOT_OK	OK	NOT_OK	x	N\A
DVD_EB_TEST_RESULT	OK	OK	OK	x	NOT_OK	N\A
DVD Test result	OK	Partially OK	Partially OK	NOT_OK	NOT_OK	ABORTED

Table 2: DVD test result

3.1.1.3 Driver´s Vigilance Device Isolation

3.1.1.3.1 Function design

In case of DVD failure, the DVD itself applies the emergency brake (fail safe) therefore in this case there shall be the possibility to isolate the device from the EB loop to be able to move the train: the DVD can be isolated through a sealed switch.

3.1.1.3.1.1 Normal condition

In case of failure of the DVD the driver isolates it from the brakes.

3.1.1.3.1.2 Failure cases

A. DVD is not isolated from the brake when the isolation switch is in isolation position
B. DVD is isolated from the brake when the isolation switch is in normal position

3.1.1.3.1.3 Failure detection

A. Failure is self-evident. The train remains braked
B. Failure can be detected during the daily test initiated by the driver. This failure however can be expected only after switch movement from isolation to normal position, therefore after the action the execution of the test is expected.

3.1.1.3.1.4 Failure handling

A. To release brake application pneumatic valve could be bypassed
B. The failure shall be repaired during maintenance

3.1.1.3.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.322
Description	It shall be possible to Bypass the EB applied by the DVD in case of a faulty DVD system.
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.184	A sealed switch on the cab shall be foreseen to bypass the DVD brake application in the emergency brake loop.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_09.02-Zefiro-ES.CONCEPT.185	The TCMS shall acquire the status of the DVD isolation Switch though a digital input.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.646	If TCMS detect incoherence between the readbacks status of DVD isolation switch (in primary and redundant sections), it shall set an EVENT	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.02-Zefiro-ES.CONCEPT.186	The TCMS shall warn the driver through the IDU when the DVD is isolated in the active cab.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.02-Zefiro-ES.CONCEPT.187	Status of DVD Isolation Switch shall be provided to JRU	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.188	If TCMS detects the isolation of DVD function, it shall inhibit DVD test (refer to 2F_09.02.Zefiro-Europe.TRS.254)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.574	TCMS shall inform driver/maintainer staff when the DVD inhibition selector is actuated in the other CAB.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

3.1.2 Automatic Train Protection System (ATP)

3.1.2.1 Visualization

3.1.2.1.1 Function design

3.1.2.1.1.1 Normal condition

In case of ETCS supervision, the speed is normally shown on the ETCS DMI.
In case of ETCS system failure and consequent isolation through CEA, it is possible to run only with ASFA signalling. In this condition, the ASFA shall visualize the speed on ASFA display.
In case of ASFA supervision the speed is normally shown on ASFA display.
In case ASFA system failure and consequent isolation (through Combinador General), is NOT possible to run only with ETCS (in level 0). Emergency brake is applied.
In case of ATP system failure (both ETCS and ASFA) and consequent isolation (both ETCS and ASFA), the speed will be displayed on the IDU.

3.1.2.1.1.2 Failure cases

A. ATP (both ETCS and ASFA) system wrongly detected as isolated by the TCMS => speed is displayed contemporaneously on ATP DMI (the one active at the moment) and on IDU
B. ATP (both ETCS and ASFA) system wrongly detected as not isolated by the TCMS => speed signal is not available to the driver

3.1.2.1.1.3 Failure detection

A. Failure self-evident
B. Failure self-evident

3.1.2.1.1.4 Failure handling

A. Failure isolated and repaired during maintenance
B. Failure isolated and repaired during maintenance

3.1.2.1.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.323
Description	The train speed shall be displayed on the diagnostic display in case of absence of the ATP systems (both ETCS and ASFA).
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.194	If the ATP is isolated through: - by the isolation switch (CEA) for ETCS and by the isolation switch (ASFA Combinador general) for ASFA the train speed shall be displayed on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_09.02-Zefiro-ES.CONCEPT.195	The TCMS shall acquire the status of the ETCS isolation switch (CEA) and of ASFA isolation switch (ASFA Combinador general). Refer to RD 1sheet 26 and 27.	Derived Requirement	4S_09.02.09.-.Conventional Train Control – Circuits		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.196	The TCMS shall warn the driver through the IDU when the ETCS ATP is isolated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.02-Zefiro-ES.CONCEPT.197	The TCMS shall warn the driver through the IDU when the ASFA ATP is isolated (ASFA BRK_INHIBIT contacts are both closed).	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.198	When the CEA is operated, a lamp on driver desk shall be commanded.	Derived Requirement	4S_09.02.09.-.Conventional Train Control – Circuits		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.199	The status of the CEA switch shall be provided to the TELOC/JRU for recording purpose.	Derived Requirement	4S_09.02.09.-.Conventional Train Control – Circuits		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.200	The status of the ASFA isolation switch (ASFA Combinador general) shall be provided to the TELOC/JRU for recording purpose. (ETCS message 255 – Proprietary data)	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.644	TCMS shall acquire ASFA_BRK_INHIBIT contacts and set an event if an incoherence is detected.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

ID	2F_09.02.Zefiro-Europe.TRS.324
Description	The JRU system shall provide the time to the train for synchronization to all equipment onboard.
Safety level	SIL 0

The time for on-board synchronization will come from TELOC/JRU and be sent to TCMS. TCMS will use the MCG via SNTP to do the synchronization.

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.203	The TELOC/JRU shall send the time in UTF format to TCMS for time synchronization on-board.	Derived Requirement	4S_09.06.06.-.Data Recorder - Juridical Recorder		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.204	The TCMS shall use the time from the TELOC/JRU for on-board time synchronization	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.1.2.2 ATP Direction control

3.1.2.2.1 Function design

The ATP monitor the direction selected by the driver and the actual train movement: in case of incoherence it applies the emergency brake.

3.1.2.2.1.1 Normal condition

The selected direction is neutral (no selection), the train moves forward or backward => ATP emergency brake application
The selected direction is forward, the train moves backward => ATP emergency brake application
The selected direction is backward, the train moves forward => ATP emergency brake application

3.1.2.2.1.2 Failure cases

A. Loss of ATP emergency brake application

3.1.2.2.1.3 Failure detection

A. Failure detected during system power on self-test

3.1.2.2.1.4 Failure handling

A. Failure isolated and repaired during maintenance

3.1.2.2.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.325
Description	The requirements of signals necessary for ATP to control the driving direction of the train during ATP system supervision shall be defined. The defined driving directions could be: - Forward Direction - Backward Direction
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.211	The train shall provide to the ATP the signal position Forward of the direction controller through two digital inputs	Derived Requirement	4S_09.02.09.-.Conventional Train Control – Circuits		SIL 2
2F_09.02-Zefiro-ES.CONCEPT.212	The train shall provide to the ATP the signal position Backward of the direction controller through two digital inputs	Derived Requirement	4S_09.02.09.-.Conventional Train Control – Circuits		SIL 2
2F_09.02-Zefiro-ES.CONCEPT.213	The ATP shall read the information received by the train in accordance to the following logic: Forward | Backward | Meaning 0 | 0 | Neutral 0 | 1 | Backward 1 | 0 | Forward 1 | 1 | Error→ neutral	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 2

3.1.2.3 ATP Isolation

3.1.2.3.1 Function design

In case of ETCS failure the ETCS itself applies the emergency brake (fail safe) therefore in this case there shall be the possibility to isolate the device to be able to move the train: the ETCS can be isolated through a CEA switch.
In this condition, the train can run only with ASFA signalling. In case of ASFA failure, ASFA itself applies the emergency brake (fail safe) therefore in this case there shall be the possibility to isolate the device to be able to move the train: the ATP can be isolated through a ASFA Combinador general switch.
If only ASFA signalling is in failure, after its isolation through ASFA Combinador General, the train can moves using only ETCS equipment.

3.1.2.3.1.1 Normal condition

In case of failure of the ATP (both ETCS and ASFA) the driver isolates it from the brakes.
For ETCS isolation CEA switch shall be operated and pneumatic plate shall be de-inserted in order to move the train.
For ASFA isolation, the Combinador general switch shall be operated in order to move the train.

3.1.2.3.1.2 Failure cases

A. ATP (ETCS and ASFA) is not isolated from the brake when the isolation switch is in isolation position
B. ATP (ETCS and ASFA) is isolated from the brake when the isolation switch is in normal position

3.1.2.3.1.3 Failure detection

A. Failure is self-evident. The train remains braked
B. Failure can be detected during power on test. This failure however can be expected only after switch movement from isolation to normal position, therefore after the action the execution of the test is expected.

3.1.2.3.1.4 Failure handling

A. To release brake application by ETCS, pneumatic valve could be bypassed, and to release the brake application by ASFA, Combinador general shall be operated
B. The failure shall be repaired during maintenance

3.1.2.3.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.120
Description	ATP isolation shall be possible using a sealed isolation switch.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.647	IF TCMS detect incoherence between the CEA's contacts, it shall set an EVENT	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

ID	2F_09.02.Zefiro-Europe.TRS.326
Description	It shall be possible to isolate ASFA.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.669	If the MCB connected to ASFA combinador General (Inhibition of ASFA) is open, TCMS shall set an event	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.02-Zefiro-ES.CONCEPT.670	If TCMS detect that ASFA is not Active in the Active cab, it shall set an EVENT.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

Refer to requirement in paragraph § 3.1.2.1.2.

ID	2F_09.02.Zefiro-Europe.TRS.327
Description	It shall be possible to isolate LZB via ATP.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.671	If the MCB of LZB is open, TCMS shall set an event	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.02-Zefiro-ES.CONCEPT.672	If the contactor of LZB's power supply is open, TCMS shall set an Event	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

3.1.2.4 ATP Start up Mission during Cab activation

3.1.2.4.1 Function design

3.1.2.4.1.1 Normal condition

The system shall propose to the driver the start of mission when the cab is activated

3.1.2.4.1.2 Failure cases

A. ATP does not detect the cab activation=> the mission shall be cancelled

3.1.2.4.1.3 Failure detection

A. Failure is self-evident

3.1.2.4.1.4 Failure handling

A. Failure isolated and repaired during maintenance

3.1.2.4.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.328
Description	ATP shall be able to manage cab. Active Signal Information which allows the ATP to start its Mission.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.228	The status " cab Active shall be provided to the ETCS ATP through two antivalent digital input.	Derived Requirement	4S_09.02.09.-.Conventional Train Control – Circuits		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.673	ASFA ATP shall be powered ON/OFF according to the status of the cabin (activated/de-activated)	Derived Requirement	4S_09.02.09.-.Conventional Train Control – Circuits		SIL0
2F_09.02-Zefiro-ES.CONCEPT.674	If TCMS detect that ASFA is Active in the not Active cab, it shall set an EVENT.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

3.1.2.5 ATP Emergency Brake control

3.1.2.5.1 Function design

3.1.2.5.1.1 Normal condition

In ETCS supervision, the ETCS applies the emergency brake commanding directly the dedicated local electro valves: the brake pipe is then discharged from the active cab

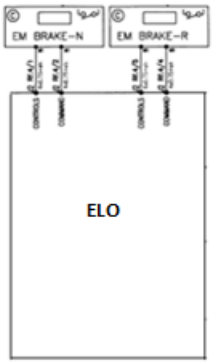


Figure 4: ELO – Pneumatic plates interface

In parallel to the pneumatic plate command the ATP commands also

- the inhibit brake pipe refill command is used to keep the pipe empty, during the emergency brake. This command is applied, de-energizing the emergency loop.
- The auxiliary emergency brake loop output (this contact is used to de-energized the loop) – OPTION not used in this moment

In ASFA supervision, the ASFA applies the emergency brake commanding the de energization of the emergency brake loop.

- In the figure below is represented a logical scheme of the signal used by ELO/ASFA to manage ASFA EB application in accordance to the Table 3:
    - ASFA_brake_inhibit is constituted by 2 NC contacts inside the Combinador general, if ASFA is isolated /powered OFF the asfa_brake_inhibit contacts, are closed to bypass the emergency brake contacts.
      When ASFA is operating these contacts are opened.
    - ASFA_AKT1 and ASFA_AKT2 are used by ELO to avoid ASFA braking during ETCS supervision.
      When the train is ETCS supervised, these contacts are closed by ELO.
      When ASFA is in charge of supervision, ELO shall open the contacts.
      ELO read backs the status of these contacts.
    - ASFA_brake_CMD is the contact the has the functionality to apply the emergency brake in ASFA supervision.
      ASFA opens the contact to apply the emergency brake.

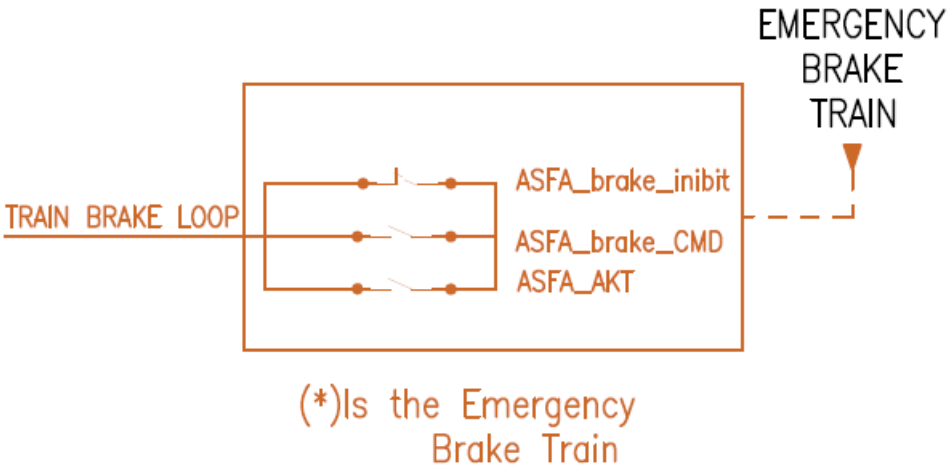


Figure 5: Logical emergency brake interface

ASFA_BRAKE_INHIBIT	ASFA_BRK_CDM	ASFA_AKT	EB by ASFA
1	x	x	Not applied
0	0	0	Applied
x	1	x	Not applied
x	x	1	Not applied

Table 3: ASFA_EB application conditions

3.1.2.5.1.2 Failure cases

A. (ETCS) Loss to de energize the emergency brake loop => the brake pipe will be discharged only from the active cab therefore the equivalent time of the brake application will be greater
B. (ASFA) Loss to de-energization of the brake loop => The train is not stopped when needed.

3.1.2.5.1.3 Failure detection

A. Failure detected by the TCMS
B. Failure detected by the TCMS

3.1.2.5.1.4 Failure handling

A. Failure isolated and repaired during maintenance
B. Failure isolated and repaired during maintenance

3.1.2.5.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.329
Description	Every ATP equipment shall be integrated in the train's electrical and pneumatical circuits in such a manner that it can order Emergency Brake.
Safety level	SIL 4

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.240	The ETCS will order emergency brake commanding its dedicated pneumatic plate	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 4
2F_09.02-Zefiro-ES.CONCEPT.705	STM LZB sends via FFIS interface, the EB command to ETCS, and ETCS apply it.	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 4
2F_09.02-Zefiro-ES.CONCEPT.241	The inhibit brake pipe refill command (Refer to RD 1 sheet 25) shall be applied at the same time as the EB_command to the pneumatic plate.	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.242	When the ETCS command the inhibit brake pipe refill, Emergency brake shall be applied in the whole train by de-energizing the Emergency brake loop .	Derived Requirement	4S_09.02.09.-.Conventional Train Control – Circuits		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.648	When TCMS detects an incoherence between the status of the "inhibit brake pipe refill command" and the status of the urgency loop, it shall set an EVENT	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.02-Zefiro-ES.CONCEPT.243	If ETCS orders Emergency brake an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.02-Zefiro-ES.CONCEPT.244	TCMS shall set an event when ASFA request EB (ASFA_BRK_CMD=0), the emergency loop is de-energized and no EB is requested by ETCS ATP (pipe_refill_inhibition =1). Note: considering these triggering conditions, the event is set when ASFA really request the EB and when ETCS is in supervision and AKT_1 or AKT_2 relay is in fault but not when ASFA run in external mode in normal conditions.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.02-Zefiro-ES.CONCEPT.245	TCMS shall detect which pneumatic plate is inserted.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.246	TCMS Shall provide to the JRU the status of the pneumatic plate for recording purpose.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.675	If the MCB of ASFA brake command is open, TCMS shall set an event	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

ID	2F_09.02.Zefiro-Europe.TRS.330
Description	ASFA shall be able to command an Emergency Braking.
Safety level	SIL 4

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.248	The ASFA applies the emergency brake commanding the de energization of the emergency brake loop accordingly to the Figure 5	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 4
2F_09.02-Zefiro-ES.CONCEPT.249	TCMS shall acquire the status of ASFA_BRK_CMD relays contacts and set and event if an incoherence is detected.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

ID	2F_09.02.Zefiro-Europe.TRS.331
Description	LZB shall be able via ATP to command an Emergency Braking.
Safety level	SIL 4

ID	2F_09.02.Zefiro-Europe.TRS.182
Description	In parallel with the pneumatic Emergency brake order to the Braking system an electrical signal shall be sent to the Braking system to apply Emergency brake
Safety level	SIL 3

The requirement 2F_09.02.Zefiro-Europe.TRS.182 will not be considered for this release of the SFRS

3.1.2.6 ATP Service Brake control

3.1.2.6.1 Function design

3.1.2.6.1.1 Normal condition

The ATP will command the service brake when needed (First Line of Intervention), the TCMS will forward the request to the brake system that shall apply the **100%** of service brake.
Note: in case the service brake command will not be applied or in case the speed reduction is not enough to avoid crossing the second line of intervention the ATP will command the emergency brake.

3.1.2.6.1.2 Failure cases

A. Service brake not applied when commanded by the ATP

3.1.2.6.1.3 Failure detection

A. Failure self-evident: EB applied by the ATP

3.1.2.6.1.4 Failure handling

A. Failure isolated and repaired during maintenance

3.1.2.6.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.132
Description	ATP shall be able to order Service Brake.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.259	The ETCS shall command the full service brake through MVB.	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.706	STM LZB sends via FFIS interface, the SB command to ETCS, and ETCS forward it as per 2F_09.02-Zefiro-ES.CONCEPT.259	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.260	The TCMS shall apply full service brake when ordered by ATP, according to Brake blending rules.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.1.2.7 Brake pressure

3.1.2.7.1 Function design

3.1.2.7.1.1 Normal condition

The brake pressure information shall be provided to the ATP for braking model uses.

3.1.2.7.1.2 Failure Cases

N/A

3.1.2.7.1.3 Failure Detection

N/A

3.1.2.7.1.4 Failure Handling

N/A

3.1.2.7.2 Requirements

-

ID	2F_09.02.Zefiro-Europe.TRS.333
Description	ATP shall be able to manage Brake Pressure Information.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.274	The information of AVERAGE BCP across TRAILER axles (without ED), shall be used for brake_pressure information exchanged between TCMS and ATP.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.1.2.8 Traction Cut-off

3.1.2.8.1 Function design

3.1.2.8.1.1 Normal condition

The ATP will command the SW traction block when needed (phase and voltage transition, speed limitation), the TCMS will forward the request to the traction system that shall remove the traction effort.
The TCO command shall be managed, at train level, in a safe way when emergency brake is applied (traction cut off opening the safe traction line) in the other cases, like overpassing the warning speed, is requested a traction bloc that it has NO SIL target.

3.1.2.8.1.2 Failure cases

A. Traction effort wrongly not removed => the MCB remains closed with traction effort

3.1.2.8.1.3 Failure detection

A. Failure not detected

3.1.2.8.1.4 Failure handling

A. Failure isolated and repaired during maintenance

3.1.2.8.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.334
Description	ATP shall be able to command a traction cut-off in the whole train.
Safety level	SIL 2

<table>
<tr><th>Nr</th><th>Description</th><th>Type</th><th>Derived to</th><th>User Interface</th><th>SIL</th></tr>
<tr><td>2F_09.02-Zefiro-ES.CONCEPT.281</td><td>The ETCS shall command traction cut off when necessary through a digital output. Refer to RD 1 sheet 25</td><td>Derived Requirement</td><td>4S_09.06.-.Automatic Train Protection Systems</td><td></td><td>SIL 1</td></tr>
<tr><td>2F_09.02-Zefiro-ES.CONCEPT.282</td><td>The ETCS shall send to TCMS (MVB) the traction cut off command</td><td>Derived Requirement</td><td>4S_09.06.-.Automatic Train Protection Systems</td><td></td><td>SIL 2</td></tr>
<tr><td>2F_09.02-Zefiro-ES.CONCEPT.283</td><td>TCMS shall acquire the traction block command (both digital and MVB command) and apply TCO as follow:

<table><tr><th>O_TC1_C</th><th>OBU_TR_TCO_Cmd</th><th>Meaning</th></tr><tr><td>1</td><td>1</td><td>Traction cut-off not commanded</td></tr><tr><td>0</td><td>1</td><td>Traction cut-off commanded.</td></tr><tr><td>1</td><td>0</td><td>Traction cut-off commanded.</td></tr><tr><td>0</td><td>0</td><td>Traction cut-off commanded.</td></tr></table>
The TCO command shall be managed, at train level, in a safe way when emergency brake is applied (traction cut off opening the safe traction line) in the other cases, like overpassing the warning speed, is requested a traction block that it has NO SIL target.</td><td>Derived Requirement</td><td>4S_09.03.05.-.TCMS Software</td><td></td><td>SIL 2</td></tr>
<tr><td>2F_09.02-Zefiro-ES.CONCEPT.649</td><td>The TCO command (digital output and MVB) shall be applied by ETCS during braking (EB and SB).</td><td>Derived Requirement</td><td>4S_09.06.-.Automatic Train Protection Systems</td><td></td><td>SIL 2</td></tr>
<tr><td>2F_09.02-Zefiro-ES.CONCEPT.707</td><td>STM LZB sends via FFIS interface, the TCO command to ETCS, and ETCS forward it as per 2F_09.02-Zefiro-ES.CONCEPT.283.</td><td>Derived Requirement</td><td>4S_09.06.-.Automatic Train Protection Systems</td><td></td><td>SIL 0</td></tr>
<tr><td>2F_09.02-Zefiro-ES.CONCEPT.284</td><td>If ATP (ETCS+STM LZB) traction block command is applied AND the switch CEA is not operated then Traction block shall be applied in the whole train.</td><td>Derived Requirement</td><td>4S_09.03.05.-.TCMS Software</td><td></td><td>SIL 0</td></tr>
<tr><td>2F_09.02-Zefiro-ES.CONCEPT.285</td><td>If the CEA is operated, the traction block command by the isolated ATP (ETCS+STM LZB) shall be bypassed.</td><td>Derived Requirement</td><td>4S_09.03.05.-.TCMS Software</td><td></td><td>SIL 0</td></tr>
<tr><td>2F_09.02-Zefiro-ES.CONCEPT.650</td><td>TCMS shall set an event when incongruence between normal and reserve TCO line is detected.</td><td>Derived Requirement</td><td>4S_09.03.05.-.TCMS Software</td><td>EVENT</td><td>SIL 0</td></tr>
</table>

NOTE: The TCO command shall be managed, at train level, in a safe way when emergency brake is applied (traction cut off opening the safe traction line) in the other cases, like overpassing the warning speed, is requested a traction bloc that it has NO SIL target.

3.1.2.9 Main Circuit breaker command

3.1.2.9.1 Function design

3.1.2.9.1.1 Normal condition

The ATP, will send to the TCMS the signals (both for ERTMS trackside orders and STM orders), for MCB management, according to RD 6

3.1.2.9.1.2 Failure cases

3.1.2.9.1.3 Failure detection

3.1.2.9.1.4 Failure handling

3.1.2.9.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.335
Description	The ATP shall give additional information to manage the opening or closing of the main circuit breaker.
Safety level	SIL 0

The management of MCB is demanded to TCMS, that for BL3 specification shall manage automatically the MCB, as described in 3.1.2.13 or 3.1.2.14

3.1.2.10 Phase Transition

3.1.2.10.1 Function design

3.1.2.10.1.1 Normal condition

3.1.2.10.1.2 Failure cases

3.1.2.10.1.3 Failure detection

3.1.2.10.1.4 Failure handling

3.1.2.10.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.336
Description	ATP (ETCS) shall command the Entry/Exit to the location of change phase.
Safety level	SIL 0

The phase transition sequence is now managed via ETCS track condition, in particular is reported in 3.1.2.14.

3.1.2.11 Voltage Transition

3.1.2.11.1 Function design

3.1.2.11.1.1 Normal condition

3.1.2.11.1.2 Failure cases

3.1.2.11.1.3 Failure detection

3.1.2.11.1.4 Failure handling

3.1.2.11.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.337
Description	ATP (ETCS) shall command the Entry/Exit to the location of change of traction system.
Safety level	SIL 0

The voltage transition sequence is now managed via ETCS track condition, in particular is reported in 3.1.2.12 and 3.1.2.13.

3.1.2.12 Change of Traction System

3.1.2.12.1 Function design

3.1.2.12.1.1 Normal condition

The change of Traction system sequence is represented in the image below.
Trackside information are received by the ATP and then are transmitted to the train.
In practice the change of traction system Track Condition may be used combined with another Track Condition e.g. a powerless section with pantograph to be lowered (3.1.2.13).

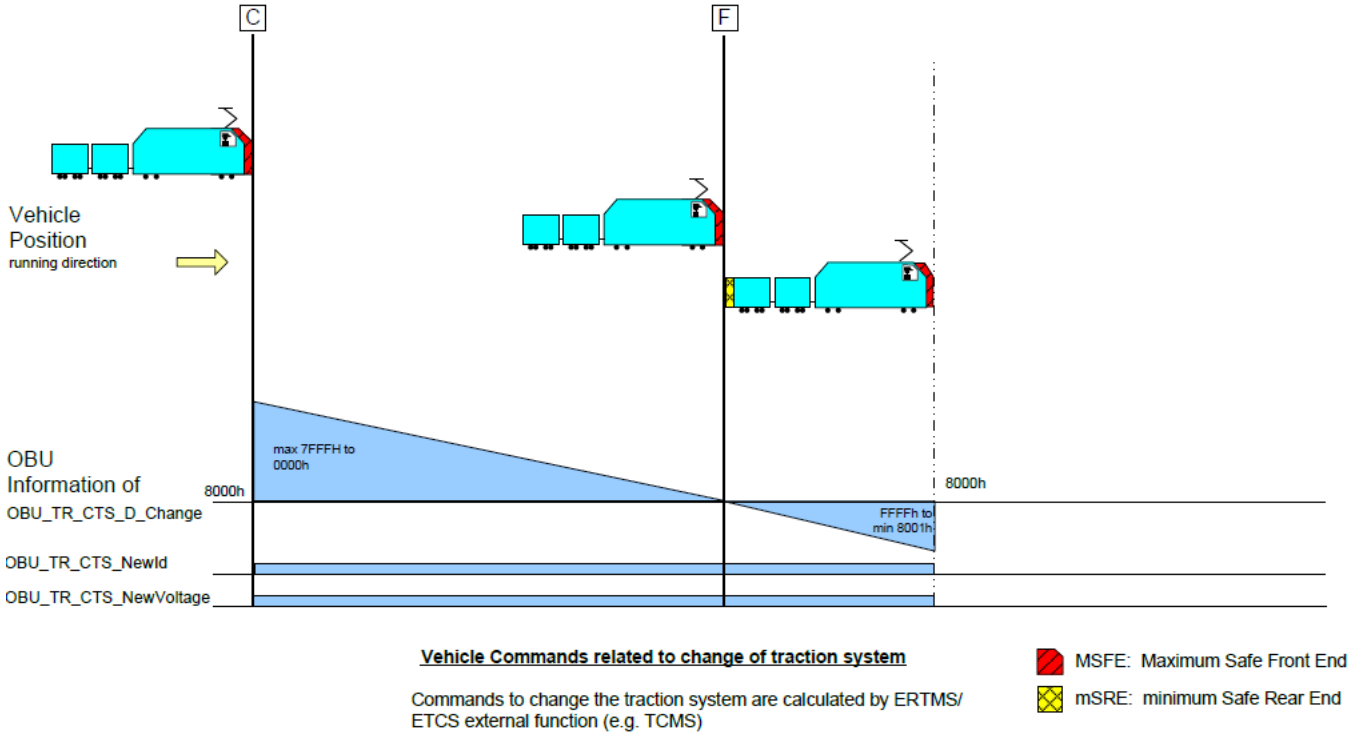


Figure 6: change of traction system sequence description

3.1.2.12.1.2 Failure cases

3.1.2.12.1.3 Failure detection

3.1.2.12.1.4 Failure handling

3.1.2.12.2 Requirements

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.303	The TCMS shall acquire the signals about Change of traction system in accordance to RD 6 as follow: OBU_TR_CTS_D_Change (Distance to new traction system) OBU_TR_CTS_NewId (The new traction system (NID_CTRACTION defined in subset026 cap 7)) OBU_TR_CTS_NewVoltage (The new voltage (M_VOLTAGE)).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.575	The ATP shall provide the information about new voltage system and distance to the TCMS through MVB using the following signals: OBU_TR_CTS_D_Change OBU_TR_CTS_NewId OBU_TR_CTS_NewVoltage	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.576	Deleted				
2F_09.02-Zefiro-ES.CONCEPT.577	When the OBU_TR_CTS_D_Change is expired (0x8000h) a timer of 30 s shall be started, if the condition to rise the Panto are not met until the timer expires, the automatic sequence shall be aborted (POC/PCF icon on IDU shall be activated) and the driver is now in charge to perform manual action to restore the traction of the train.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.319	If the line voltage system selected by the driver does not correspond to the line voltage system reported by the ATP (OBU_TR_CTS_NewVoltage) an event shall be set, and the pantograph shall not be raised.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

Note: NID_CTRACTION is defined in SUBSET-026-7 v360 at § 7.5.1.86.1.
M_VOLTAGE is defined in SUBSET-026-7 v360 at § 7.5.1.78.

3.1.2.13 Powerless Section with Pantograph to be Lowered – Trackside orders

3.1.2.13.1 Function design

3.1.2.13.1.1 Normal condition

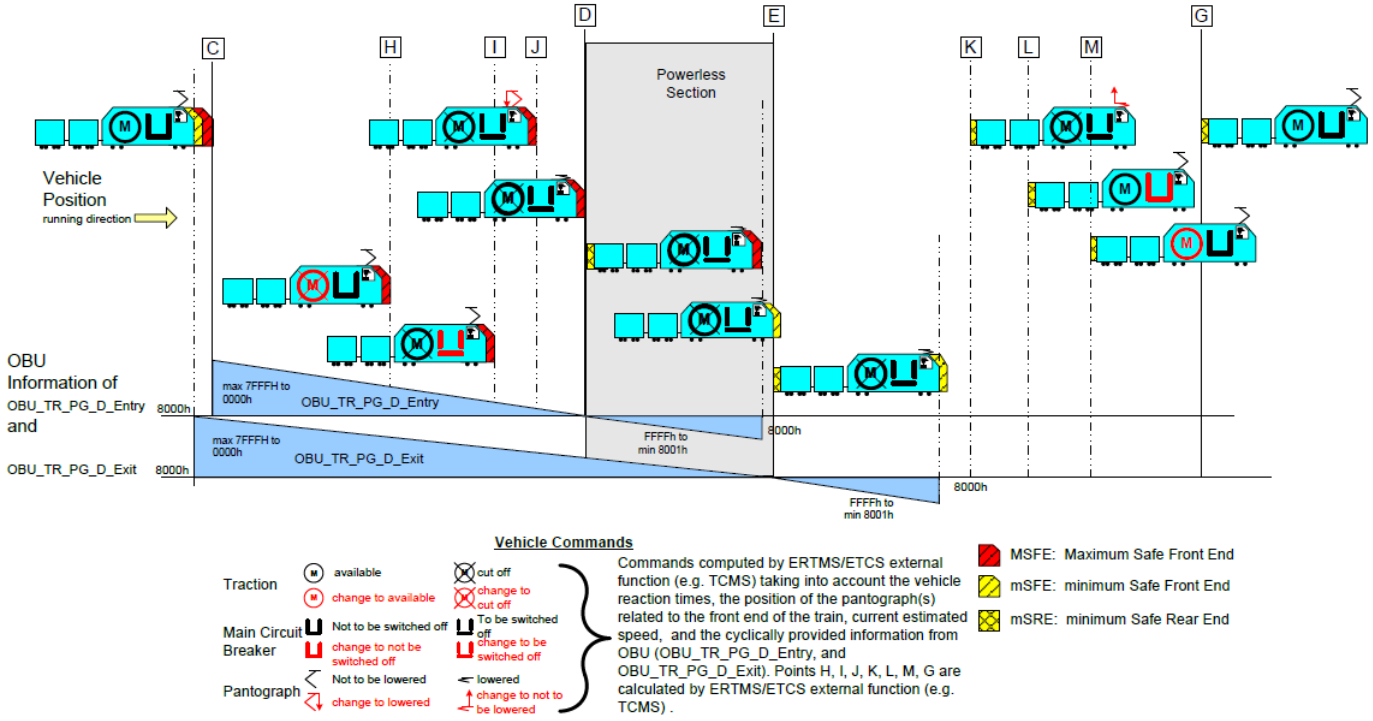


Figure 7: 3.1.2.13 Powerless Section with Pantograph to be Lowered/voltage transition sequence description

ERTMS/ETCS on-board equipment shall provide the remaining distance to the start location point D (OBU_TR_PG_D_Entry) and the remaining distance to the end location point E (OBU_TR_PG_D_Exit) to the TCMS.
TCMS commands to cut off the traction when the max safe front end of the train reaches the point H.
TCMS commands to switch off the main circuit breaker when the max safe front end of the train reaches the point I.
TCMS commands to lower the pantograph when the max safe front end of the train reaches the point J.
TCMS commands the pantograph not to be lowered the when the min safe rear end of the train reaches the point K.
TCMS commands the main circuit not to be switched off when the min safe rear end of the train reaches the point L.
TCMS commands to change the traction to be available when the min safe rear end of the train reaches the point M.
When the min safe rear end of the train reaches the point G the traction is available.
Note: The train could also control each pantograph separately and therefore react according to the location of the various pantographs, not according to the location of the train front/rear end as assumed in the figure.
About STM LZB, the tracks in which it is installed are electrified with 25kV (AC) system, so no panto management by LZB is foresee in service. The panto STM order are foresee in this project as predisposition in order to be compliant to RD 6 and STM LZB specifications.

3.1.2.13.1.2 Failure cases

3.1.2.13.1.3 Failure detection

3.1.2.13.1.4 Failure handling

3.1.2.13.2 Requirements

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.314	The TCMS shall acquire the signals about Panto management in case of Powerless Section with Pantograph to be Lowered – Trackside orders is received by the ATP in accordance to RD 6 as follow: OBU_TR_PG_D_Entry OBU_TR_PG_D_Exit	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.315	The TCMS shall acquire the signals about Panto management in case of Pantograph – STM Orders is received by the ATP in accordance to RD 6 as follow: O_PG_C / OBU_TR_PG_Cmd with the meaning: 0: Lower Pantograph 1: Raise pantograph	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02-Zefiro-ES.CONCEPT.316	Starting from the moment when TCMS receives the information from ATP, it shall perform all the action in order to lower the pantograph in the proper way, before that OBU_TR_PG_D_Entry value reach 0x0000h.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.317	The TCMS shall raise the Panto, only when the OBU_TR_PG_D_Exit is expired (0x8000h), the voltage selected by the driver is coherent with the value received in OBU_TR_CTS_NewVoltage and other functional conditions are met.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.578	TCMS shall use the following parameters to manage the neutral section entering sequence: - OBU_TR_PG_D_Entry: remaining entry distance - T_P: time needed to lower the panto - T_MCB: time needed to open MCB - T_TCO: time needed to reach null traction current - V= train speed	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.579	TCMS commands to cut off the traction when the max safe front end of the train reaches the point H: H= OBU_TR_PG_D_Entry-(T_P+T_MCB+T_TCO)*v	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.580	TCMS commands to open the main circuit breaker when the max safe front end of the train reaches the point I I= OBU_TR_PG_D_Entry-(T_P+T_MCB)*v	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.581	TCMS commands to lower the pantograph when the max safe front end of the train reaches the point J J= OBU_TR_PG_D_Entry-(T_P)*v	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.318	When the pantograph has been raised the TCMS shall close the MCB,and restore the traction only if all functional condition are met.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.582	When TCMS receives O_PG_C / OBU_TR_PG_Cmd = 0, it shall start the sequence to lower the panto. In order: 1. Command traction block 2. Open MPS when all related functional conditions are met. 3. Lower the panto when all related functional conditions are met.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02-Zefiro-ES.CONCEPT.583	When TCMS receive O_PG_C / OBU_TR_PG_Cmd = 1 (neutral section exit), the TCMS shall apply a delay of x m before star processing it, and then it shall start the sequence to restore the traction as before neutral section. In order: 1. Raise the panto 2. Close MPS if all related functional conditions are met 3. Restore the traction. x= - 180 m in single configuration - 385 m in multiple configuration	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02-Zefiro-ES.CONCEPT.584	THE ATP shall provide the information about remaining distance and exit distance of the neutral section through MVB using the following signals: OBU_TR_PG_D_Entry OBU_TR_PG_D_Exit	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.585	THE ATP shall provide the information about stm orders for panto through MVB using the following signal: O_PG_C / OBU_TR_PG_Cmd	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0

Note: requirements 2F_09.02-Zefiro-ES.CONCEPT.582 and 2F_09.02-Zefiro-ES.CONCEPT.583 are to be intended as mitigation because, as explained in RD 9, no change of traction system is managed by LZB in the lines:

- 010. MADRID-PUERTA DE ATOCHA a SEVILLA-SANTA JUSTA.
- 030. BIF. MÁLAGA-A.V. a MÁLAGA-MARÍA ZAMBRANO.

3.1.2.14 Powerless Section with Main Power Switch to be Switched Off – Trackside Orders

3.1.2.14.1 Function design

3.1.2.14.1.1 Normal condition

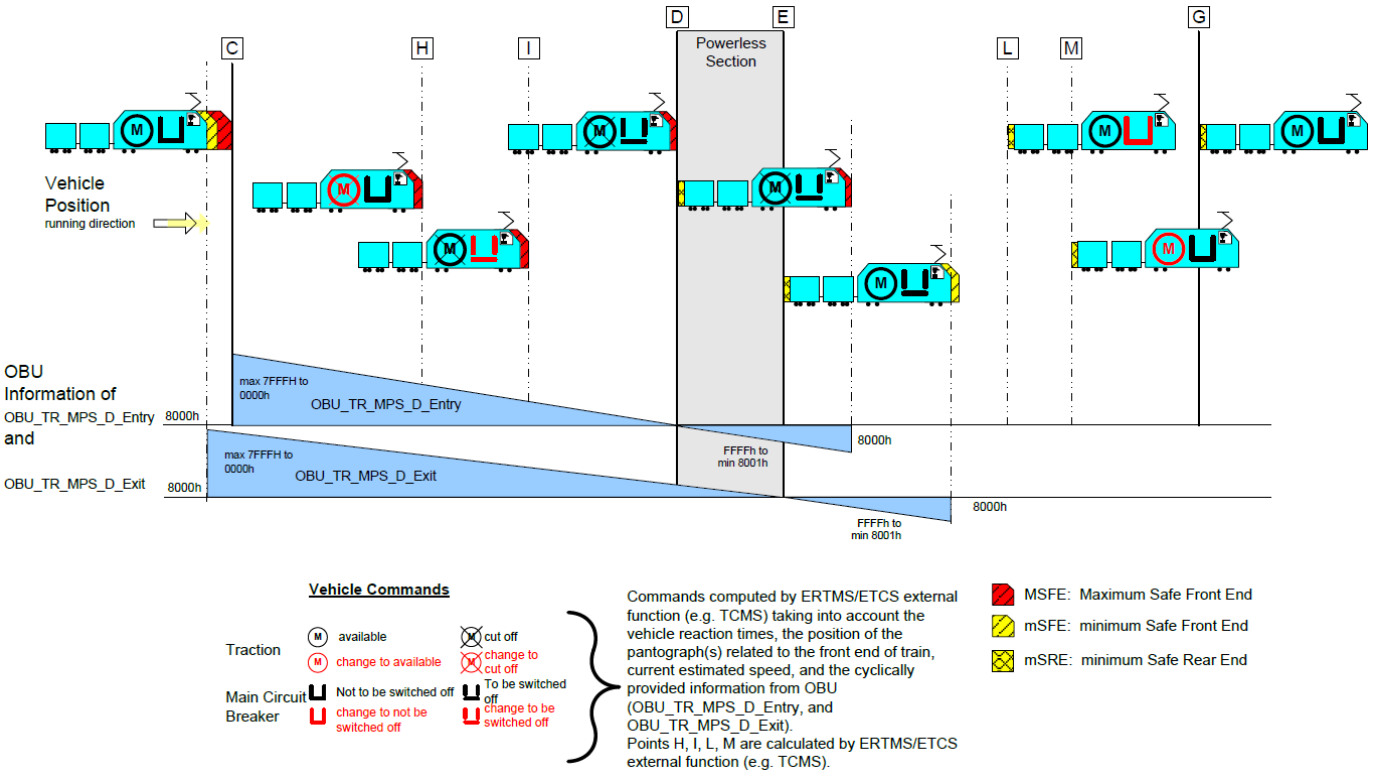


Figure 8: 3.1.2.14 Powerless Section with Main Power Switch to be Switched Off/ phase transition sequence description

ERTMS/ETCS on-board equipment shall provide the remaining distance to the start location point D (OBU_TR_MPS_D_Entry) and the remaining distance to the end location point E (OBU_TR_MPS_D_Exit) to the TCMS.
TCMS commands to cut off the traction before the max safe front end of the train reaches the point H.
TCMS commands to switch off the main circuit breaker before the max safe front end of the train reaches the point I.
TCMS commands the main circuit not to be switched off when the min safe rear end of the train reaches the point L.
TCMS commands to change the traction to be available when the min safe rear end of the train reaches the point M.
When the min safe rear end of the train reaches the point G the traction is available.

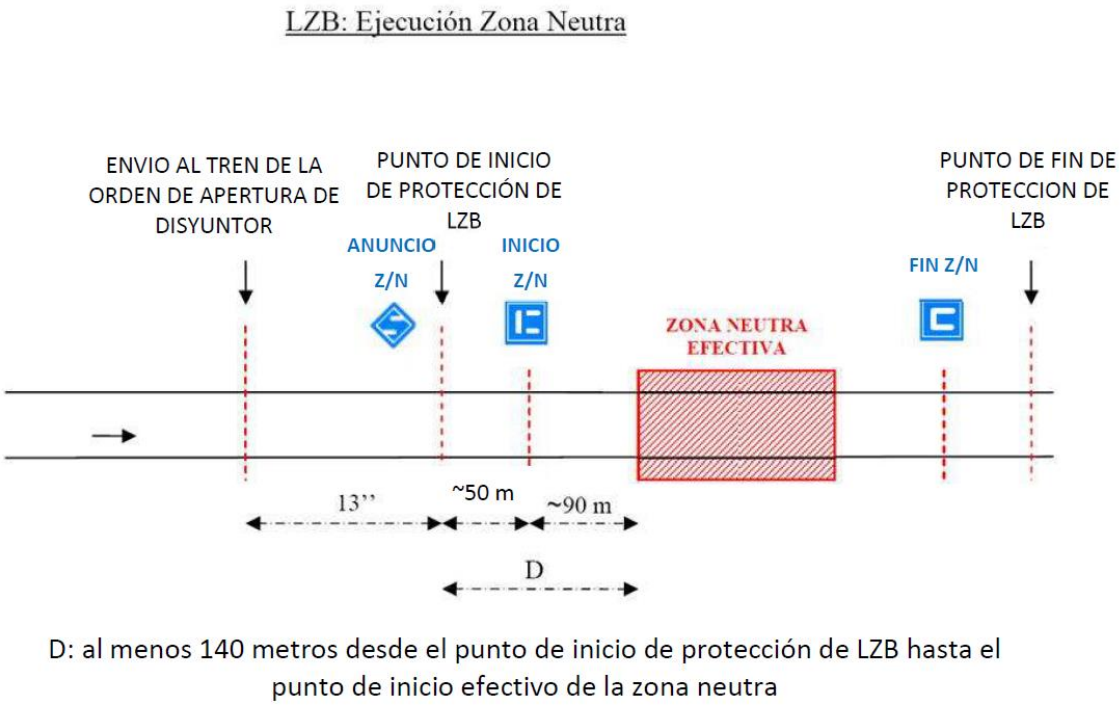


Figure 9: LZB way side management, for powerless section as per RD 9

3.1.2.14.1.2 Failure cases

3.1.2.14.1.3 Failure detection

3.1.2.14.1.4 Failure handling

3.1.2.14.2 Requirements

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.329	The TCMS shall acquire the signals about MCB management in case of Powerless Section with Main Power Switch to be Switched Off – Trackside Orders is received by ATP, in accordance to RD 6 as follow: OBU_TR_MPS_D_Entry (remaining distance from entering neutral section) OBU_TR_MPS_D_Exit (remaining distance to the end of neutral section)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.586	The ATP shall provide the information about remaining distance and exit distance of the neutral section through MVB using the following signals: OBU_TR_MPS_D_Entry OBU_TR_MPS_D_Exit	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.330	The TCMS shall acquire the signals about MCB management in case of Main Power Switch – STM Orders received by ATP in accordance to RD 6 as follow: O_MPS_C / OBU_TR_MPS_Cmd Meaning: 0: Main Power Switch to be switched off 1: Main Power Switch NOT to be switched off	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.653	When TCMS receive O_MPS_C / OBU_TR_MPS_Cmd=0, the TCMS shall start the sequence to open the MPS. In order: 1. Command traction block 2. Open MPS when all related functional conditions are met.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.654	When TCMS receive O_MPS_C / OBU_TR_MPS_Cmd=1 (neutral section exit), the TCMS shall start the sequence to restore the traction as before neutral section. In order: 1. Close MPS if all related functional conditions are met 2. Restore the traction.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.587	THE ATP shall provide the information about remaining distance and exit distance of the neutral section through MVB using the following signal: O_MPS_C / OBU_TR_MPS_Cmd	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.331	Starting from the moment that TCMS receives the OBU_TR_MPS_D_Entry, it shall perform all the action needed to open the MPS (MCB) before OBU_TR_MPS_D_Entry reach 0x0000h.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.588	TCMS shall use the following parameters to manage the neutral section entering sequence: - OBU_TR_PG_D_Entry: remaining entry distance - T_MCB: time needed to open MCB - T_TCO: time needed to reach null traction current - V= train speed	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.589	TCMS commands to cut off the traction when the max safe front end of the train reaches the point H: H= OBU_TR_PG_D_Entry-(T_MCB+T_TCO)*v	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.590	TCMS commands to open the main circuit breaker when the max safe front end of the train reaches the point I I= OBU_TR_PG_D_Entry-(T_MCB)*v	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.591	When the OBU_TR_MPS_D_Exit is expired (0x8000h) a timer of 30 s shall be started, if the condition to close MCB are not met until the timer expires, the automatic sequence shall be aborted (POC/PCF icon on IDU shall be activated) and the driver is now in charge to perform manual action to restore the traction of the train.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.592	When O_MPS_C / OBU_TR_MPS_Cmd = 0 is received, the TCMS shall cut off the traction and then it shall open the MPS (MCB) only if all related functional condition are met.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.593	When O_MPS_C / OBU_TR_MPS_Cmd = 1 is received the TCMS shall apply a delay of x m before star processing it, and then it shall close the MPS (MCB) if all functional conditions are met. x= - 180 m in single configuration - 385 m in multiple configuration	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.655	When ETCS or LZB request to open MCB with (MPS track condition or STM order), TCMS shall set an EVENT.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.02-Zefiro-ES.CONCEPT.656	During phase separation (neutral) section, TCMS shall set an event	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.657	If TCMS detects one condition for which to forbid automatic POC or PCF sequence, it shall interrupt the sequence (if the sequence is running) and show on IDU the “PCF/POC unavailable” icon.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL0
2F_09.02-Zefiro-ES.CONCEPT.658	When TCMS shows the “POC/PCF unavailable” icon, it shall set an EVENT.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

3.1.2.15 Air Tightness

3.1.2.15.1 Function design

3.1.2.15.1.1 Normal condition

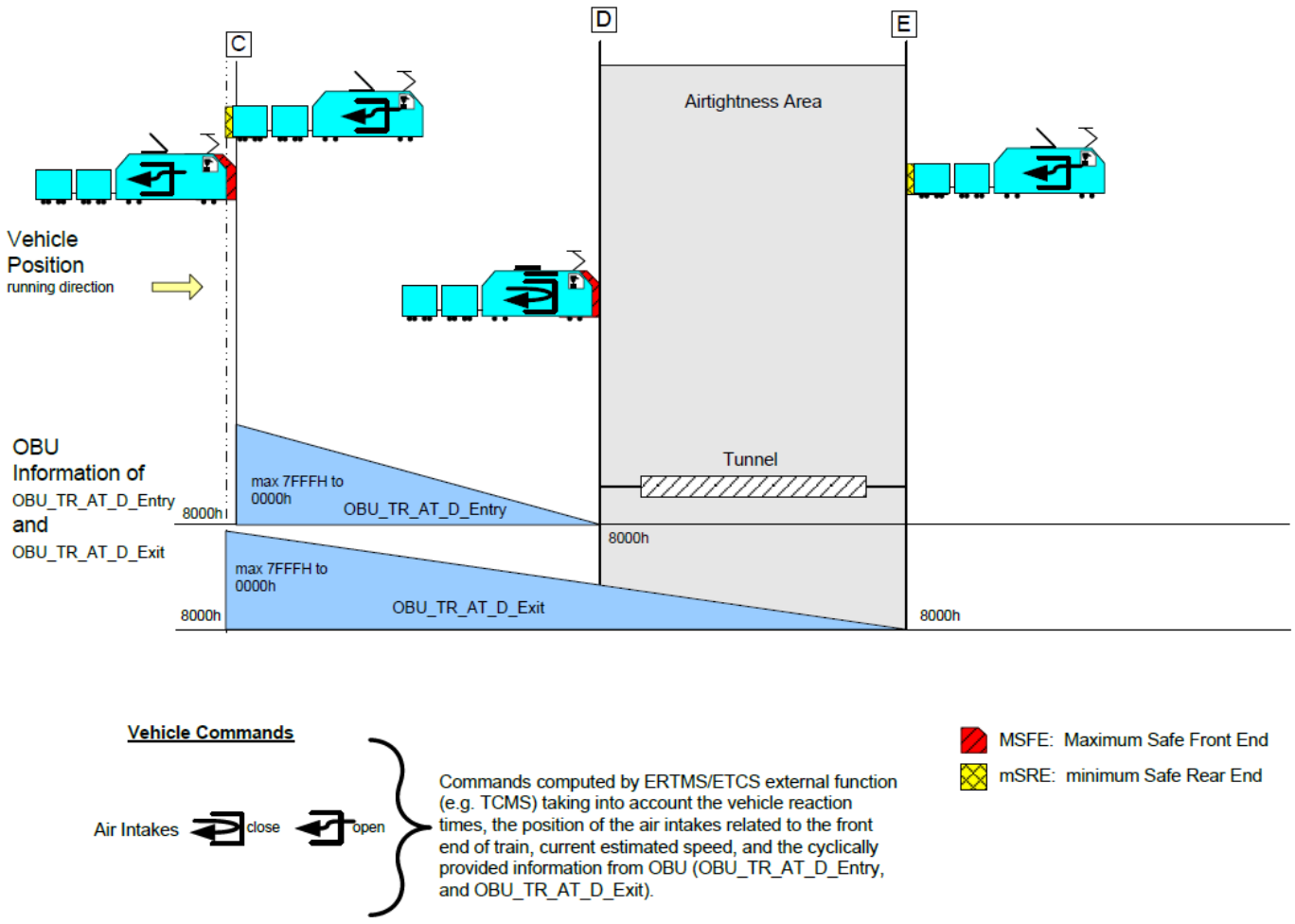


Figure 10: air tightness/Tunnel mode management

The air tightness area or tunnel area action are performed automatically in FS.
ERTMS/ETCS on-board equipment shall provide the remaining distance to the start location point D (OBU_TR_AT_D_Entry) and the remaining distance to the end location point E (OBU_TR_AT_D_Exit) to the TCMS.
See § 5.18.6 of SUBSET-026-5 v360

3.1.2.15.1.2 Failure cases

3.1.2.15.1.3 Failure detection

3.1.2.15.1.4 Failure handling

3.1.2.15.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.285
Description	ATP shall command the Tunnel Entry/Exit section.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.342	The TCMS shall acquire the signals about air tightness management in case of Air Tightness Area – Trackside orders is received by the ATP in accordance to RD 6 as follow: OBU_TR_AT_D_Entry OBU_TR_AT_D_Exit	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.594	THE ATP shall provide the information about remaining distance and exit distance of the Tunnel area through MVB using the following signals: OBU_TR_AT_D_Entry OBU_TR_AT_D_Exit	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.343	The TCMS shall acquire the signals about Panto management in case of Air Tightness – STM Orders is received by the ATP in accordance to RD 6 as follow: OBU_TR_AT_Cmd/O_AT_C with the meaning: 0: Open air conditioning intake 1: Close air conditioning intake	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.595	THE ATP shall provide the information about STM command for tunnel area through MVB using the following signal: OBU_TR_AT_Cmd/O_AT_C	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.596	TCMS shall close the dumpers before OBU_TR_AT_D_Entry = 0x0000h.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.597	TCMS shall use the following parameters to manage the tunnel area entering sequence: - OBU_TR_AT_D_Entry: remaining entry distance - T_AT: time needed to close the dumpers - V= train speed	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.598	TCMS commands to close the dumpers when the max safe front end of the train reaches the point D: D= OBU_TR_AT_D_Entry-(T_AT)*v	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.599	TCMS shall open the dumpers when the OBU_TR_AT_D_Exit is expired (0x8000h).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.600	TCMS shall close the dumpers when receives OBU_TR_AT_Cmd=1	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.601	When TCMS receives OBU_TR_AT_Cmd=0, it shall apply a delay of x m before start processing it, and then it shall open the dumpers x= - 180 m in single configuration - 385 m in multiple configuration	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.1.2.16 LZB button

3.1.2.16.1 Function design

3.1.2.16.1.1 Normal condition

The STM LZB can read the status of its buttons through “LZB NVS Extended Train interface”: TCMS acquire the status of each button (antivalent contacts) and sent it via MVB to ETCS that internally forward the message to STM LZB through STM interface.

3.1.2.16.1.2 Failure cases

3.1.2.16.1.3 Failure detection

3.1.2.16.1.4 Failure handling

3.1.2.16.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.339
Description	LZB shall acquire the status of its buttons via TCMS (through ATP).
Safety level	SIL 0

For details see: §8.4 Packet STM-203: LZB buttons in ESSILS19940D301 – STM LZB ES Interface Spec_v0.4
The LZB NVS Extended Train interface is addressed to the On-Board SAP = 18. This is the LZB-specific function for LZB Button’s status and the LZB-ATO interface (in the same physical address than DMI Cab A Main)

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.354	TCMS shall acquire and send to the ATP the ALERTAR LZB button: TasteWachsam1FS1: “Alertar“ button. 1=actuated TasteWachsam2FS1: “Alertar“ button (antivalent). 0=actuated as defined in RD 8	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.355	TCMS shall acquire and send to the ATP the LIBERAR LZB button: TasteFrei1FS1: “Liberar“ button. 1=actuated TasteFrei2FS1: “Liberar“ button (antivalent). 0=actuated as defined in RD 8	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.356	TCMS shall acquire and send to the ATP the REBASAR LZB button: TasteBefehl1FS1: “Rebasar“ button. 1=actuated TasteBefehl2FS1: “Rebasar“ button (antivalent). 0=actuated as defined in RD 8	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.357	TCMS shall acquire and send to the ATP the EMERGENCIA OTRA VIA LZB: Nothalt1FS1: “Emergencia otra vía“ button. 1=actuated Nothalt2FS1: “Emergencia otra vía“ button (antivalent). 0=actuated as defined in RD 8	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.358	TCMS shall acquire and send to the ATP the FALLO LZB button: LZBStoer1: “Rec. Avería LZB“ switch. 1=actuated LZBStoer2: “Rec. Avería LZB“ switch (antivalent). 0=actuated as defined in RD 8	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.359	FALLO LZB state shall be provided to JRU for recording purpose.	Derived Requirement	4S_09.02.09.-.Conventional Train Control – Circuits		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.602	The TCMS shall raise an event if detect the incoherence between the antivalent contacts of the ALERTAR button.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.02-Zefiro-ES.CONCEPT.603	The TCMS shall raise an event if detect the incoherence between the antivalent contacts of the LIBERAR button.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.02-Zefiro-ES.CONCEPT.604	The TCMS shall raise an event if detect the incoherence between the antivalent contacts of the REBASAR button.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.02-Zefiro-ES.CONCEPT.605	The TCMS shall raise an event if detect the incoherence between the antivalent contacts of the EMERGENCIA OTRA VIA button.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.02-Zefiro-ES.CONCEPT.606	The TCMS shall raise an event if detect the incoherence between the antivalent contacts of the FALLO LZB button.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.02-Zefiro-ES.CONCEPT.607	If an incoherence is detected in each button, the relative validity bit in TR Telegram 3 shall be set to 0.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.1.2.17 LZB TRANS LAMP

3.1.2.17.1 Function design

The LZB TRANS Lamp shall be managed by ELO during LZB→ETCS, LZB→ASFA and reverse transitions.

3.1.2.17.1.1 Normal condition

N/A

3.1.2.17.1.2 Failure Cases

N/A

3.1.2.17.1.3 Failure Detection

N/A

3.1.2.17.1.4 Failure Handling

N/A

3.1.2.17.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.341
Description	ETCS shall manage LZB-TRANS lamp on driver desk.
Safety level	SIL 4

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.373	The ETCS shall command LZB trans lamp in accordance to RD 1 pag 73	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 4

3.1.2.18 Sleeping

3.1.2.18.1 Function design

Sleeping functionality **is not implemented** at this moment for VZES project.
For this functionality the ATP shall receive 2 antivalent-contacts **for future uses**:

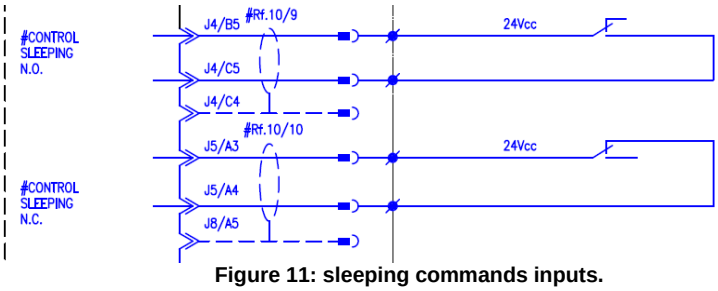


Figure 11: sleeping commands inputs.

NB: when the function will be implemented, some changes at train level will be required in order to manage properly this information.

3.1.2.18.1.1 Normal condition

The meaning of the contacts is:

NO contact	NC contact	Sleeping command
Open	Closed	Sleeping not Requested
Closed	Open	Sleeping Requested
Open	Open	Error
Closed	Closed	Error

Table 4: sleeping command inputs meaning

3.1.2.18.1.2 Failure Cases

N/A

3.1.2.18.1.3 Failure Detection

N/A

3.1.2.18.1.4 Failure Handling

N/A

3.1.2.18.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.343
Description	ATP shall be able to manage Sleeping Information (predisposition).
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.390	The sleeping command shall be provided to the ATP with the meaning: SLEEPING_COMMAND_NO: OPEN: Sleeping not Requested CLOSED: Sleeping Requested	Derived Requirement	4S_09.02.09.-.Conventional Train Control – Circuits		SIL1
2F_09.02-Zefiro-ES.CONCEPT.391	The sleeping command shall be provided to the ATP with the meaning: SLEEPING_COMMAND_NC: OPEN: Sleeping Requested CLOSED: Sleeping not Requested	Derived Requirement	4S_09.02.09.-.Conventional Train Control – Circuits		SIL1

3.1.2.19 Train Integrity

3.1.2.19.1 Function design

Train integrity functionality is not implemented at this moment for VZES project.
For this functionality the ATP shall receive 2 antivalent-contacts **for future uses**:

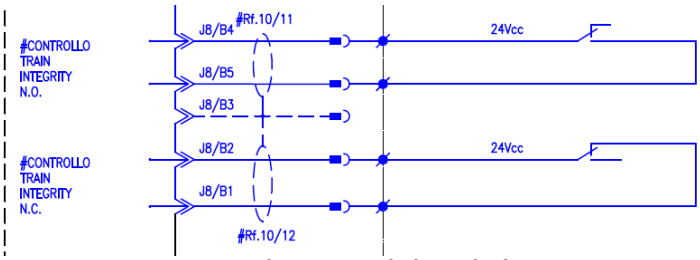


Figure 12: train integrity inputs

NB: when the function will be implemented, some changes at train level will be required in order to manage properly this information.

3.1.2.19.1.1 Normal condition

The meaning of the contacts is:

NO contact	NC contact	Train integrity input
Open	Closed	Train Integrity NOT OK
Closed	Open	Train Integrity OK
Open	Open	Error/ Train Integrity NOT OK
Closed	Closed	Error/ Train Integrity NOT OK

Table 5: train integrity inputs meaning

3.1.2.19.1.2 Failure Cases

N/A

3.1.2.19.1.3 Failure Detection

N/A

3.1.2.19.1.4 Failure Handling

N/A

3.1.2.19.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.345
Description	ATP shall be able to manage Train Integrity Information (predisposition).
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.408	The Train Integrity Status shall be provided to the ATP with the meaning: TRAIN_INTEGRITY_NO: OPEN: TRAIN INTEGRITY NOT OK CLOSED: TRAIN INTEGRITY OK	Derived Requirement	4S_09.02.09.-.Conventional Train Control – Circuits		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.409	The Train Integrity Status shall be provided to the ATP with the meaning: TRAIN_INTEGRITY _NC: OPEN: TRAIN INTEGRITY OK CLOSED: TRAIN INTEGRITY NOT OK	Derived Requirement	4S_09.02.09.-.Conventional Train Control – Circuits		SIL 0

3.1.2.20 Level Transition

3.1.2.20.1 Function design

Level transition between ETCS, ASFA and LZB shall be performed in accordance to Subset 026 and to FN-27 section in "RESOLUCIÓN CIRCULAR 3/2019" as far as Level transitions involving Level NTC (ASFA) are concerned.

During level Transition some signal are used between ELO equipment and ASFA equipment, in particular:
- ASFA_LINEA_ON: to inform ELO that ASFA is powered ON
- ASFA_AKT1 and ASFA_AKT_2 described in § 3.1.2.5
- ASFA_CONNECTED

During ETCS supervision ELO commands ASFA_CONNECTED= 0 (closed contact on ECP side) and ASFA_AKT_1 and ASFA_AKT_2= 1 (contact in emergency brake loop closed) to avoid the emergency brake applied by ASFA.

When a transition ETCS→ASFA is required, at the border ELO commands ASFA_CONNECTED=1 (contact is opened on ECP side) and after 500 ms ASFA_AKT_1,2 are opened to allow ASFA to brake during its supervision.

When a transition ASFA→ETCS is required, ASFA_Connected=0 and ASFA_AKT_1,2 contact are closed at the same time.

For safety reason ASFA_CONNECTED, ASFA_AKT_1 And ASFA_AKT_2 output shall be read-back. The ELO's outputs are connected each one to a relays in order to use one contact to check the status of the contact and the other one to operate in the circuitry.

The feedback signals are named:
ASFA_CONNECTED_FEEDBACK
ASFA_AKT_1_FEEDBACK
ASFA_AKT_2_FEEDBACK
See RD 1 page 25,26 and 74

3.1.2.20.1.1 Normal condition

The technical description and specification are demanded to ATP signalling system

3.1.2.20.1.2 Failure Cases

Analysed at ATP signalling level

3.1.2.20.1.3 Failure Detection

Analysed at ATP signalling level

3.1.2.20.1.4 Failure Handling

Analysed at ATP signalling level

3.1.2.20.2 Requirements

Analysed at ATP signalling level

3.1.2.21 Train configuration

3.1.2.21.1 Function design

The train configuration take in account the performance of the vehicle in case of nominal or degraded condition in terms of isolated bogie→ percentage of braking mass/maximum speed.
As per Donor project the possible configurations take in account from nominal condition (all bogies available) up to 5 isolated bogies.

3.1.2.21.1.1 Normal condition

No bogie isolated

3.1.2.21.1.2 Failure cases

From 1 to 5 isolated bogies

3.1.2.21.1.3 Failure detection

N/A

3.1.2.21.1.4 Failure handling

Isolation of the faulty bogie/bogies and selection of the appropriated train data set.

3.1.2.21.2 Requirements

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.621	The train data entry shall be “Swhitchable type” data entry.	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 4
2F_09.02-Zefiro-ES.CONCEPT.708	The ETCS DMI shall show the button to switch from fixed to flexible data Entry with the text: “Enter Data”	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 4
2F_09.02-Zefiro-ES.CONCEPT.709	The ETCS DMI shall show the button to switch from flexible to fixed data Entry with the text: “Select Type”	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 4

<table>
<tr><th>Nr</th><th>Description</th><th>Type</th><th>Derived to</th><th>User Interface</th><th>SIL</th></tr>
<tr><td>2F_09.02-Zefiro-ES.CONCEPT.622</td><td>The ATP shall foresee at least the following dataset, in order to allow the driver to select the correct configuration during data entry. in base of the effective braking capacity of the train:
- Normal condition (no isolated bogie)
- 1 isolated bogie
- 2 isolated bogies
- 3 isolated bogies
- 4 isolated bogies

All parameters associated to the configuration above are reported in a separated document,</td><td>Derived Requirement</td><td>4S_09.06.-.Automatic Train Protection Systems</td><td></td><td>SIL 4</td></tr>
<tr><td>2F_09.02-Zefiro-ES.CONCEPT.710</td><td>The driver is allowed to modify the following data if flexible data entry is selected:

<table><tr><th>Input Field</th><th>Keyboard</th></tr><tr><td>Train category</td><td>Dedicated keyboard</td></tr><tr><td>Length (m)</td><td>Numeric</td></tr><tr><td>Brake percentage</td><td>Numeric</td></tr><tr><td>Maximum speed (km/h)</td><td>Numeric</td></tr><tr><td>Axle load category</td><td>Dedicated keyboard</td></tr><tr><td>Airtight</td><td>Dedicated keyboard
Touch screen technology:
Key ‘7’ = No
Key ‘8’ = Yes</td></tr><tr><td>Loading gauge</td><td>Dedicated keyboard</td></tr></table></td><td>Derived Requirement</td><td>4S_09.06.-.Automatic Train Protection Systems</td><td></td><td>SIL 4</td></tr>
</table>

3.1.3 Juridical Recorder Unit (JRU)

3.1.3.1 Function design

3.1.3.1.1 Normal condition

The train recorder shall record all signal coming from the train through MVB (TCMS) or through digital input

3.1.3.1.2 Failure cases

N/A

3.1.3.1.3 Failure detection

N/A

3.1.3.1.4 Failure handling

N/A

3.1.3.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.346
Description	The JRU shall be able to storage all the Information needed at least by TSIs and Spanish NNTRs.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.433	The JRU shall be powered by permanent line	Derived Requirement	4S_09.06.06.-.Data Recorder - Juridical Recorder		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.434	The JRU shall activate recording function when the cab is activated	Derived Requirement	4S_09.06.06.-.Data Recorder - Juridical Recorder		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.435	The train shall provide to the JRU through digital inputs, at least the data defined in RD 2 annex H.3.2. The list of digital signal is in RD 7	Derived Requirement	4S_09.02.09.-.Conventional Train Control – Circuits		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.553	The TCMS shall provide to the JRU through MVB at least the data defined in RD 2 annex H.3.2. The list of MVB signal is in RD 8	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.436	The JRU shall record ETCS (inclusive STM LZB) data defined in RD 5 and received through ETHERNET	Derived Requirement	4S_09.06.06.-.Data Recorder - Juridical Recorder		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.437	THE JRU shall record the ASFA data defined in RD 3 Annex 4 and received through a serial communication.	Derived Requirement	4S_09.06.06.-.Data Recorder - Juridical Recorder		SIL 0

ID	2F_09.02.Zefiro-Europe.TRS.175
Description	A faulty JRU shall be reported to the driver.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.439	If JRU is faulty (**recording functionality unavailable**), teloc shall switch on the relative lamp on the driver's desk and send the same information through mvb (disspiavdis=1)	Derived Requirement	4S_09.02.09.-.Conventional Train Control – Circuits		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.659	When TCMS detect disspiavdis=1, it shall set an EVENT	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.02-Zefiro-ES.CONCEPT.440	When JRU memory occupation (not transmitted data) reach 75% , shall switch on the relative lamp on the driver's desk and send the same information through MVB (disspiamemfull=1)	Derived Requirement	4S_09.02.09.-.Conventional Train Control – Circuits		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.660	When TCMS detect disspiamemfull=1, it shall set an EVENT	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.02-Zefiro-ES.CONCEPT.441	If the TELOC system detect that the vigilance functionality is not available, it shall power on the relative lamp on the driver desk.	Derived Requirement	4S_09.06.06.-.Data Recorder - Juridical Recorder		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.623	When TELOC read that DVD_INHIBITION selector is actuated, the relative lamp on driver desk shall be powered on.	Derived Requirement	4S_09.06.06.-.Data Recorder - Juridical Recorder		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.624	When TELOC detect that communication with external gateway (remote download functionality) is missing, it shall send the information through MVB (disradioasAP=0) to TCMS.	Derived Requirement	4S_09.06.06.-.Data Recorder - Juridical Recorder		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.625	When TCMS detect disradioasAP=0 it shall set an event	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.02-Zefiro-ES.CONCEPT.661	when teloc detect the fault of digital cab radio (digital input on teloc side), it shall send the information through MVB (disstatradio=1)	Derived Requirement	4S_09.06.06.-.Data Recorder - Juridical Recorder		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.662	When TCMS detect disstatradio=1, it shall set an EVENT	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.02-Zefiro-ES.CONCEPT.663	When Teloc detect a fault of speed sensors it shall send this information through MVB (disstatspeedsensor=1)	Derived Requirement	4S_09.06.06.-.Data Recorder - Juridical Recorder		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.664	When TCMS detect disstatspeedsensor =1, it shall set an EVENT	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.02-Zefiro-ES.CONCEPT.665	When Teloc detect a difference of 3km/h between its sensors more than 1 minute, it shall send this information through MVB (disstatledMSV=1)	Derived Requirement	4S_09.06.06.-.Data Recorder - Juridical Recorder		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.666	When TCMS detect disstatledMSV =1, it shall set an EVENT	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.02-Zefiro-ES.CONCEPT.676	If TELOC detects internal memory error shall send this information through MVB (disstatcommCDCER=1)	Derived Requirement	4S_09.06.06.-.Data Recorder - Juridical Recorder		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.677	When TCMS detect disstatcommCDCER =1, it shall set an EVENT	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.02-Zefiro-ES.CONCEPT.678	When TELOC detects it is in a faulty condition, it shall send this information through MVB (disstatER=1)	Derived Requirement	4S_09.06.06.-.Data Recorder - Juridical Recorder		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.679	When TCMS detect disstatER =1, it shall set an EVENT	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

3.1.4 Cab Radio

3.1.4.1 Function design

3.1.4.1.1 Normal condition

The cab is equipped with a GSM-R radio for train ground voice communication. The radio will be connected to the intercom and PA system and will have proper interface to the vigilance device for the alarm generation in case of DVD intervention.
The radio shall be powered as soon as the battery contactors are closed but it shall be possible to power it also when the train is off, through a dedicated switch.

The failure of the radio will be clearly indicated on the driver desk.

3.1.4.1.2 Failure Cases

N/A

3.1.4.1.3 Failure Detection

N/A

3.1.4.1.4 Failure Handling

N/A

3.1.4.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.347
Description	The Cab Radio shall be compliant to TSIs (CCS) and Spanish NNTRs.
Safety level	SIL 0

Power management

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.448	The cab radio shall be switched on as soon as the battery are connected.	Derived Requirement	4S_09.02.09.-.Conventional Train Control – Circuits		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.449	There shall be a switch in the cabin in order to power on the cab radio when the battery are disconnected	Derived Requirement	4S_09.02.09.-.Conventional Train Control – Circuits		SIL 0

Alarm lamp

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.451	In case the cab radio is not powered or in case of cab radio major fault the cab radio shall provide this information through a contact free of voltage	Derived Requirement	4S_10.01.-.Train Radio System		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.452	When the cab radio is not available a light indication shall be switched on the driver desk	Derived Requirement	4S_09.02.09.-.Conventional Train Control – Circuits		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.453	The alarm lamp status shall be provided to the DIS/JRU	Derived Requirement	4S_09.02.09.-.Conventional Train Control – Circuits		SIL 0

Alarm message

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.455	The cab radio shall acquire the DVD_ALARM signal, in order to start the emergency call	Derived Requirement	4S_10.01.-.Train Radio System		SIL 0

Cab status

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.457	Deleted				

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.458	Deleted				

Train Running Number

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.627	Train running number shall be entered by the driver in the CAB RADIO MMI, and then it shall be shared with other subsystem via MVB	Derived Requirement	4S_10.01.-.Train Radio System		SIL 0

3.1.5 Analogic Cab Radio

The train shall be equipped with analogic cab radio system to be used where train track side communication is not covered by digital radio (voice) service.

The analogic cab radio shall be compliant to:
- UIC 751-1Julio 2002 (4.ª ed.)
- UIC 751-2 Agosto 2002 (4.ª ed.)
- UIC 751-3 Julio 2005 (4.ª ed.)

The analogic radio system shall implement an interface as per UIC 761 at least as option for public announcement form track side.

3.1.5.1 Function design

3.1.5.1.1 Normal condition

N/A

3.1.5.1.2 Failure Cases

N/A

3.1.5.1.3 Failure Detection

N/A

3.1.5.1.4 Failure Handling

N/A

3.1.5.2 Requirements

N/A

3.1.6 Paramaters

Above are described some parameter that shall be used with the same value in two or more system.

3.1.6.1 Function design

The definition of the following parameters is needed because both ETCS (BL3) and TCMS use the same parameter during the track conditions management.
According the value of the parameters, TCMS calculates the distance in which set a command and ETCS show to the driver the information.

The systems behaviour shall be aligned to avoid confusion to the driver.

Note: the parameters below have similar ones that are safety or performance relevant.
these are only used in track conditions management and can contain “margins” in order to achieve the correct behaviour in driver perception.

3.1.6.1.1 Normal condition

N/A

3.1.6.1.2 Failure Cases

N/A

3.1.6.1.3 Failure Detection

N/A

3.1.6.1.4 Failure Handling

N/A

3.1.6.2 Requirements

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02-Zefiro-ES.CONCEPT.695	T_TCO = 3 [s]	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.696	T_TCO = 3 [s]	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.697	T_MCB = 5 [s]	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.698	T_MCB = 5 [s]	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.699	T_P = 20 [s]	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.700	T_P = 20 [s]	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.701	T_AT = 4 [s]	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02-Zefiro-ES.CONCEPT.702	T_AT = 4 [s]	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0

The parameters have the following meaning:

- T_P: time needed to lower the pantograph
- T_MCB: time needed to open MCB
- T_TCO: time needed to reach null traction current
- T_AT: time needed to close the dumpers

4 Signal Interface

4.1 I/O Interface

Input (ATP):

Nr	Functional name	Source
2F_09.02-Zefiro-ES.CONCEPT.211	Forward_direction	TRAIN
2F_09.02-Zefiro-ES.CONCEPT.212	Backward_direction	TRAIN
2F_09.02-Zefiro-ES.CONCEPT.228	CAB_STATUS (antivalent contact)	TRAIN
2F_09.02-Zefiro-ES.CONCEPT.390	SLEEPING:COMMAND NO	TRAIN
2F_09.02-Zefiro-ES.CONCEPT.391	SLEEPING:COMMAND NC	TRAIN
2F_09.02-Zefiro-ES.CONCEPT.408	TRAIN_INTEGRITY_NO	TRAIN
2F_09.02-Zefiro-ES.CONCEPT.409	TRAIN_INTEGRITY_NC	TRAIN

Inputs (JRU):

Nr	Functional name	Source
2F_09.02-Zefiro-ES.CONCEPT.137	Primary chain – Safety pedal and Safety traction lever's pushbutton	TRAIN
2F_09.02-Zefiro-ES.CONCEPT.187	DVD_ISOLATION_STATUS	TRAIN
2F_09.02-Zefiro-ES.CONCEPT.199	CEA	ATP/TRAIN
2F_09.02-Zefiro-ES.CONCEPT.359	FALLO LZB	ATP
2F_09.02-Zefiro-ES.CONCEPT.435	Train variables	TCMS/TRAIN
2F_09.02-Zefiro-ES.CONCEPT.453	CAB_RADIO_FAULT_LAMP output	CAB RADIO

Input (cab radio):

Nr	Functional name	Source
2F_09.02-Zefiro-ES.CONCEPT.449	CABRADIO SWITCH	TRAIN
	CAB_STATUS to cab radio MMI	TRAIN

Input TCMS

Nr	Functional name	source
2F_09.02-Zefiro-ES.CONCEPT.150	DVD_EB_COMMAND_OUTPUT	JRU
2F_09.02-Zefiro-ES.CONCEPT.185	DVD_ISOLATION_STATUS	TRAIN
2F_09.02-Zefiro-ES.CONCEPT.172	DVD_TEST_request	IDU
2F_09.02-Zefiro-ES.CONCEPT.195	CEA	ATP/TRAIN
2F_09.02-Zefiro-ES.CONCEPT.249	ASFA_BRK_CMD	ATP
2F_09.02-Zefiro-ES.CONCEPT.245	Pneumatic_plate_1_inserted Pneumatic_plate_2_inserted	ATP

Nr	Functional name	source
2F_09.02-Zefiro-ES.CONCEPT.354	ALERTAR LZB: TasteWachsam1FS1: “Alertar“ button. 1=actuated TasteWachsam2FS1: “Alertar“ button (antivalent). 0=actuated	TRAIN
2F_09.02-Zefiro-ES.CONCEPT.355	LIBERAR LZB TasteFrei1FS1: “Liberar“ button. 1=actuated TasteFrei2FS1: “Liberar“ button (antivalent). 0=actuated	TRAIN
2F_09.02-Zefiro-ES.CONCEPT.356	REBASAR LZB TasteBefehl1FS1: “Rebasar“ button. 1=actuated TasteBefehl2FS1: “Rebasar“ button (antivalent). 0=actuated	TRAIN
2F_09.02-Zefiro-ES.CONCEPT.357	EMERGENCIA OTRA VIA LZB: Nothalt1FS1: “Emergencia otra vía“ button. 1=actuated Nothalt2FS1: “Emergencia otra vía“ button (antivalent). 0=actuated	TRAIN
2F_09.02-Zefiro-ES.CONCEPT.358	FALLO LZB LZBStoer1: “Rec. Avería LZB“ switch. 1=actuated LZBStoer2: “Rec. Avería LZB“ switch (antivalent). 0=actuated	TRAIN

Outputs (ATP):

Nr	Functional name	Destination
2F_09.02-Zefiro-ES.CONCEPT.242	Brake pipe refilling inhibition	TRAIN (EB LOOP)
2F_09.02-Zefiro-ES.CONCEPT.248	ASFA_BRAKE_COMMAND_OUTPUT	TRAIN (EB LOOP)
2F_09.02-Zefiro-ES.CONCEPT.249	ASFA_BRAKE_COMMAND_OUTPUT	TCMS
2F_09.02-Zefiro-ES.CONCEPT.245	Pneumatic_plate_1_inserted Pneumatic_plate_2_inserted	TCMS
2F_09.02.Zefiro-Europe.TRS.182	AUXILIARY_EMERGENCY_BRAKE_LOOP (not implemented in EB loop)	TRAIN (EB LOOP)
2F_09.02-Zefiro-ES.CONCEPT.281	TRUCTION_CUT_OFF	TRAIN
2F_09.02-Zefiro-ES.CONCEPT.282	TRUCTION_CUT_OFF	TCMS
	CEA SELECTOR	TRAIN

Output (JRU):

Nr	Functional name	destination
	DVD_EB_COMMAND_OUTPUT_1	TRAIN
	DVD_EB_COMMAND_OUTPUT_2	TRAIN
	DVD_EB_COMMAND_OUTPUT	TCMS
	TELOC FAULT LAMP	TRAIN
	FULL MEMORY LAMP	TRAIN
	DVD_INHIBITED_LAMP	TRAIN

Outputs (Cab Radio):

Nr	Functional name	Destination
	CAB_RADIO_FAULT_LAMP output	TRAIN
	CAB_RADIO_FAULT_LAMP output	JRU

Output TCMS

Nr	Functional name	Destination

4.2 SW Interface

Input ATP:

Nr	Functional name	Source
RD 5	ALERTAR LZB: TasteWachsam1FS1: “Alertar“ button. 1=actuated TasteWachsam2FS1: “Alertar“ button (antivalent). 0=actuated	TCMS
RD 5	LIBERAR LZB TasteFrei1FS1: “Liberar“ button. 1=actuated TasteFrei2FS1: “Liberar“ button (antivalent). 0=actuated	TCMS
RD 5	REBASAR LZB TasteBefehl1FS1: “Rebasar“ button. 1=actuated TasteBefehl2FS1: “Rebasar“ button (antivalent). 0=actuated	TCMS
RD 5	EMERGENCIA OTRA VIA LZB: Nothalt1FS1: “Emergencia otra vía“ button. 1=actuated Nothalt2FS1: “Emergencia otra vía“ button (antivalent). 0=actuated	TCMS
RD 5	FALLO LZB LZBStoer1: “Rec. Avería LZB“ switch. 1=actuated LZBStoer2: “Rec. Avería LZB“ switch (antivalent). 0=actuated	TCMS
2F_09.02-Zefiro-ES.CONCEPT.274	BRAKE_PRESSURE	TCMS
	JRU message	ATP

Input JRU:

Nr	Functional name	Source
2F_09.02-Zefiro-ES.CONCEPT.174	DVD_TEST_ACTIVATED	TCMS
2F_09.02-Zefiro-ES.CONCEPT.174	DVD_EB_TEST	TCMS
2F_09.02-Zefiro-ES.CONCEPT.437	ASFA variables	ATP
2F_09.02-Zefiro-ES.CONCEPT.200	Combinador General	ATP
	Pneumatic_plate_1_inserted Pneumatic_plate_2_inserted	TCMS

Input Cabradio:

Nr	Functional name	Source
2F_09.02-Zefiro-ES.CONCEPT.154	DVD_ALARM	JRU

output ATP

Nr	Functional name	DESTINATION
2F_09.02-Zefiro-ES.CONCEPT.259	SERVICE_BRAKE_COMMAND	TCMS
2F_09.02-Zefiro-ES.CONCEPT.283	TRUCTION_CUT_OFF	TCMS
2F_09.02-Zefiro-ES.CONCEPT.329	Powerless Section with Main Power Switch to be Switched Off – Trackside Orders OBU_TR_MPS_D_Entry OBU_TR_MPS_D_Exit	TCMS
2F_09.02-Zefiro-ES.CONCEPT.330	Main Power Switch – STM Orders O_MPS_C / OBU_TR_MPS_Cmd	TCMS
2F_09.02-Zefiro-ES.CONCEPT.303	OBU_TR_CTS_D_Change	TCMS
2F_09.02-Zefiro-ES.CONCEPT.303	OBU_TR_CTS_NewId (NID_CTRACTION)	TCMS
2F_09.02-Zefiro-ES.CONCEPT.303	OBU_TR_CTS_NewVoltage (M_VOLTAGE).	TCMS
2F_09.02-Zefiro-ES.CONCEPT.314	Powerless Section with Pantograph to be Lowered – Trackside orders OBU_TR_PG_D_Entry OBU_TR_PG_D_Exit	TCMS
2F_09.02-Zefiro-ES.CONCEPT.315	Pantograph – STM Orders O_PG_C / OBU_TR_PG_Cmd	TCMS
2F_09.02-Zefiro-ES.CONCEPT.342	Air Tightness Area – Trackside orders OBU_TR_AT_D_Entry OBU_TR_AT_D_Exit	TCMS
2F_09.02-Zefiro-ES.CONCEPT.343	Air Tightness – STM OBU_TR_AT_Cmd/O_AT_C	TCMS
2F_09.02-Zefiro-ES.CONCEPT.200	Combinador General	JRU

Output JRU:

Nr	Functional name	Destination
2F_09.02-Zefiro-ES.CONCEPT.154	DVD_ALARM	CABRADIO
2F_09.02-Zefiro-ES.CONCEPT.174	DVD_pedal_test	TCMS
2F_09.02-Zefiro-ES.CONCEPT.174	DVD_TL_button_test	TCMS
2F_09.02-Zefiro-ES.CONCEPT.174	DVD_EB_TEST_RESULT	TCMS
2F_09.02-Zefiro-ES.CONCEPT.203	TIME	TCMS
2F_09.02-Zefiro-ES.CONCEPT.439	FAULT_EVENT	ATP
2F_09.02-Zefiro-ES.CONCEPT.440	FULL_MEMORY_EVENT	ATP
2F_09.02-Zefiro-ES.CONCEPT.142	DVD visual_warning	TCMS
2F_09.02-Zefiro-ES.CONCEPT.144	DVD acoustic warning	TCMS

Output cabradio

Nr	Functional name	Destination
Train running number	Train running number	JRU

The complete MVB traffic is described in RD 8

5 References

RD	Code	Title	Source	Revision
RD 1	B60X.0100361.L03.00EN	ERTMS / BL3 - On Board Sub-System LAYOUT SPECIFIC APPLICATION ILSA V300/ZEFIRO	HRSTS	02.01
RD 2	BOE-A-2020-6804	BOE – BOLETÍN OFICIAL DEL ESTADO	MINISTERIO DE TRANSPORTES, MOVILIDADY AGENDA URBANA	
RD 3	ET 03.365.008.6	Especificaciones Técnicas y Funcionales del Sistema Embarcado ASFA Digital	ADIF	I edicion +M1+M2+ Erratum 2020
RD 4	ESSADV07020D081	STM LZB for Spain Operation	BT/ALS	Rev.04
RD 5	EC130010069	ID_ATP	HRI	02
RD 6	subset-119	Train Interface FFFIS	UNISIG	1.1.0
RD 7	JRU_signal	-	HRI	10
RD 8	3ECP413978-0101	ICD MVB ATP	BT/ALS	A
RD 9	0210624	nota tecnica lzb zznn	ADIF	1.1